



Business Process Map User Manual

Draw processes as flowcharts, manage them as versions, finalize through approval.

Company process-map system · As of 2026-09-30 · English edition

Navigate with   or click the screen edges ·  to restart

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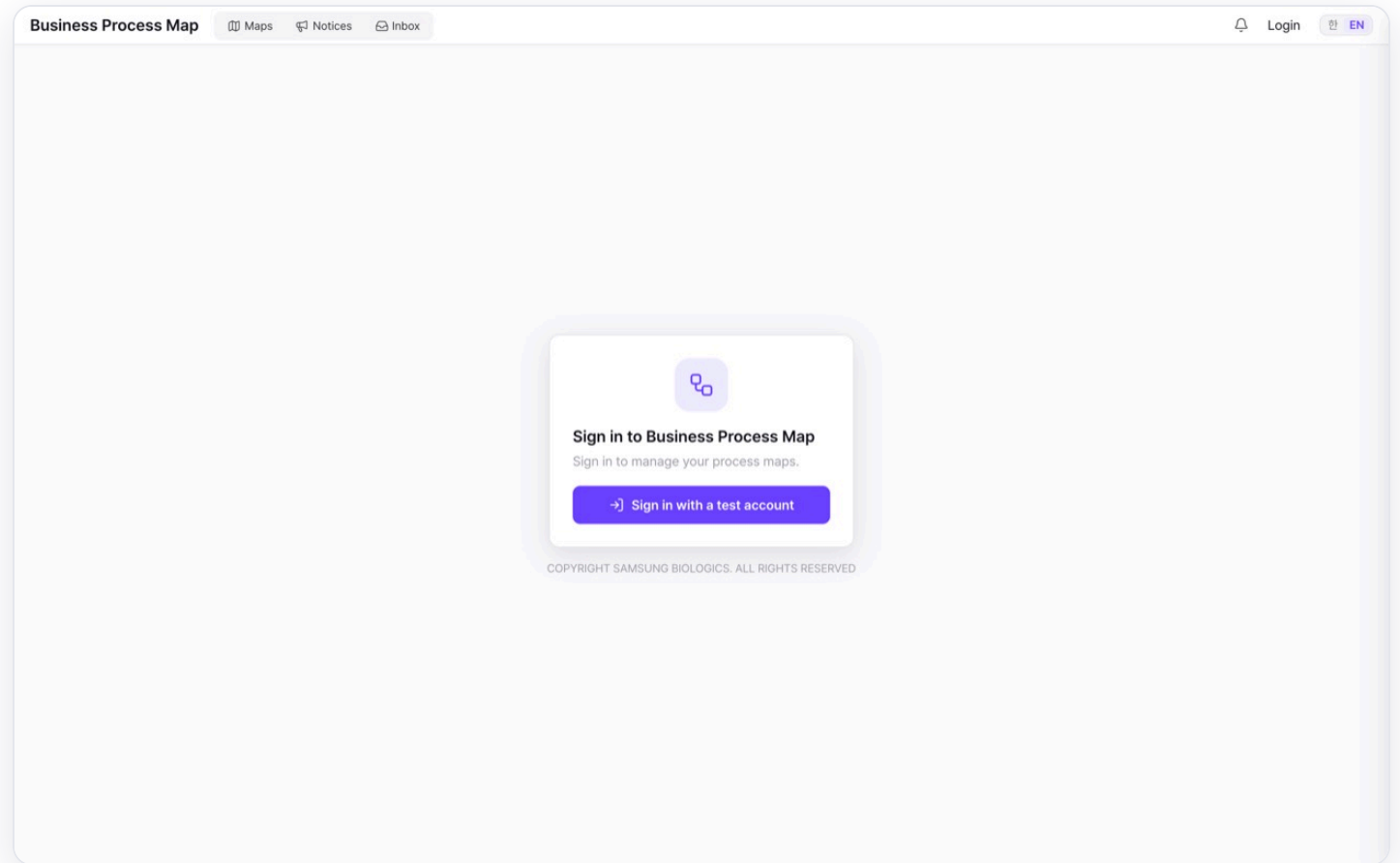
01

Getting Started

Sign in, learn the screens, and understand your role on a map.

Signing in

- **Sign in with Keycloak** — sign in with your company SSO account.
- After the first sign-in, later visits **sign you in automatically** and continue to the screen you were opening (deep link).
- Servers on company-account (LDAP) auth show an **ID/password form** instead — admin-issued local accounts work too.
- Development environments offer **Sign in with a test account**.

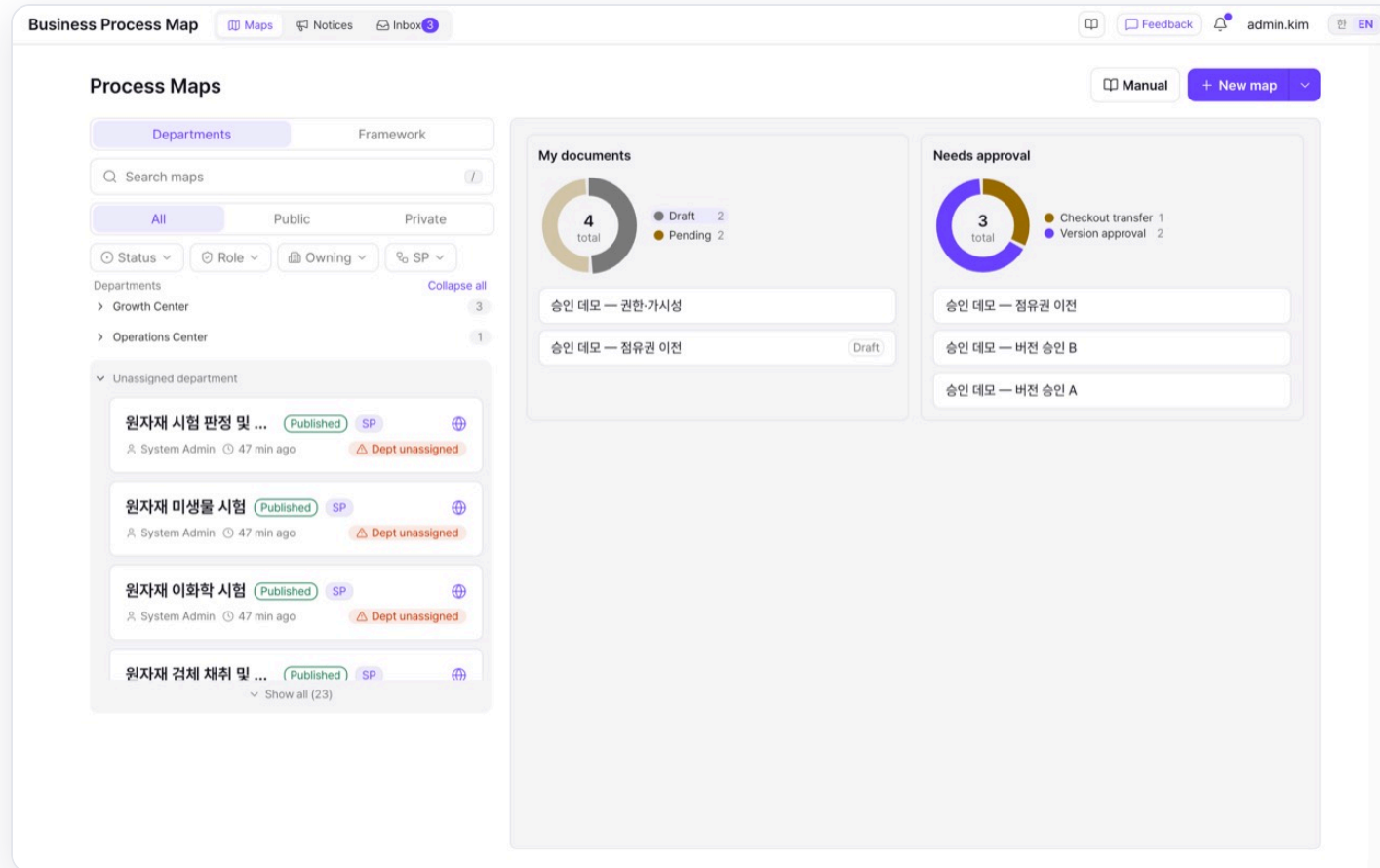


Login screen (development example)

① Getting Started

Screens at a glance

- Top tabs — Maps · Notices · Inbox.
- Top bar — Manual (KO/EN docs, search), Feedback, the bell, **Settings** (your groups · scheduled deletion · admin consoles), and the 한/EN toggle.
- Home dashboard on the right — your **personal dashboard**: profile, five activity tiles (needs my decision · my requests · checked out · unread · my feedback), and six capped sections.
- Press **/** on the map list or in the manual to jump straight to search.



Roles and visibility

- **Owner** — everything: edit, manage collaborators/approvers/visibility, delete.
- **Viewer** — read-only: browse, compare, comment.
- Every map has an **owning department**; its members automatically hold Editor access.
- The Framework adds **category admins**, separate from map roles — they edit and confirm the linkage canvases of their category and everything below (☺).
- **Editor** — edit the map and its draft versions.
- **Approver** — review and approve/reject submitted versions.
- **Public** maps are viewable by everyone; **Private** maps only by collaborators and approvers.

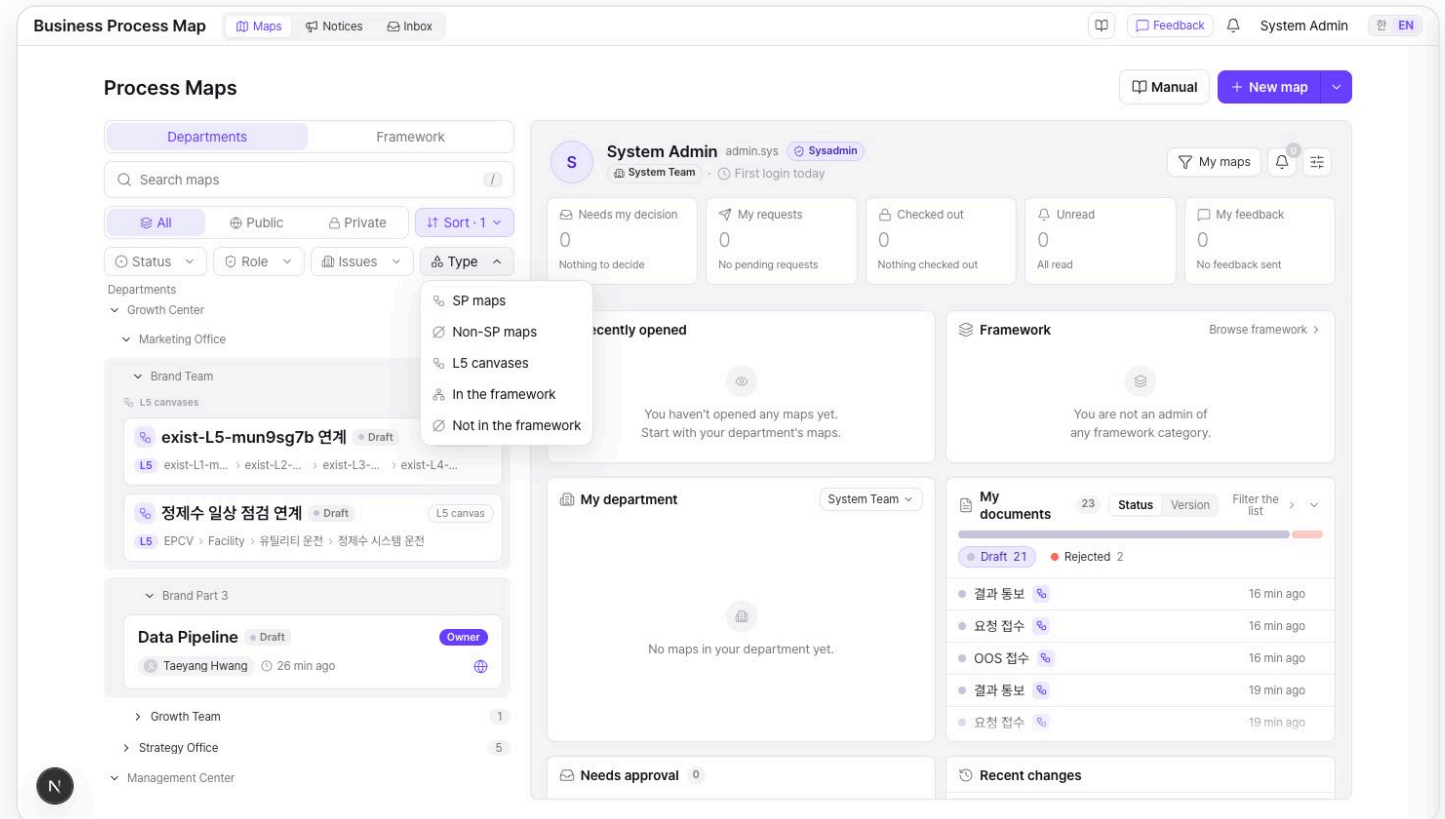
02

Home — Finding & Managing Maps

Browse maps by org tree or framework; create, copy, and manage settings.

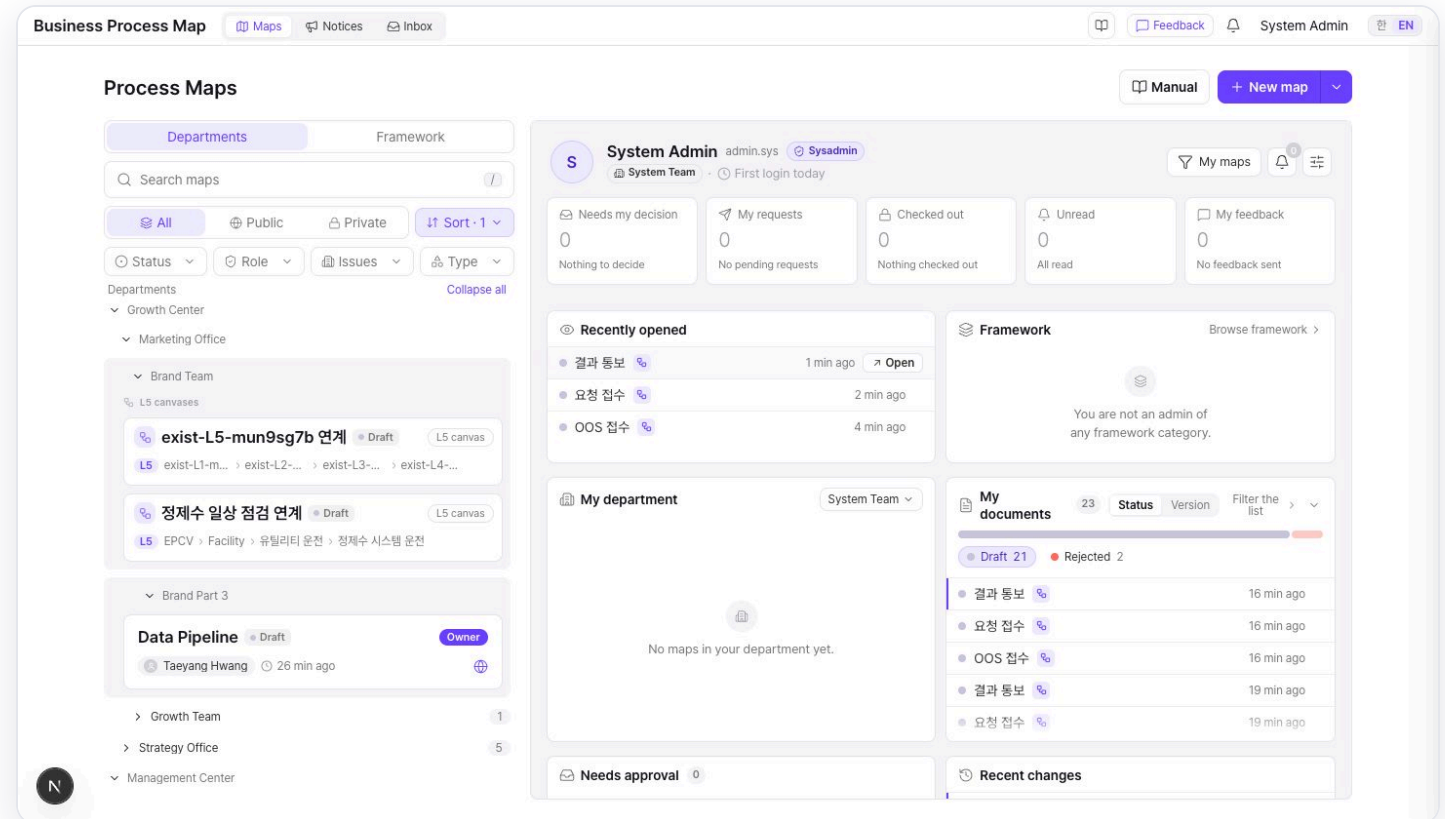
Finding maps & the detail panel

- Browse by department in the org tree — yours is pinned on top. Canvases managed by that department sit in an **L5 canvases** group; maps without one gather under Dept unassigned.
- Search (/) and **two filter rows** — All/Public/Private and **Sort** on top; Status · Role · Issues · **Type** (SP / non-SP / L5 canvases / in the framework or not) below.
- Click a card for the detail panel — owner, owning department, members, version history.
- Interview-imported maps without an owner carry an Owner unconfirmed pill until Transfer ownership.
- Panel bottom — read-only interview source notes and owner-managed Notes.
- With edit access, add/remove collaborators on the card (staged Save/Undo).



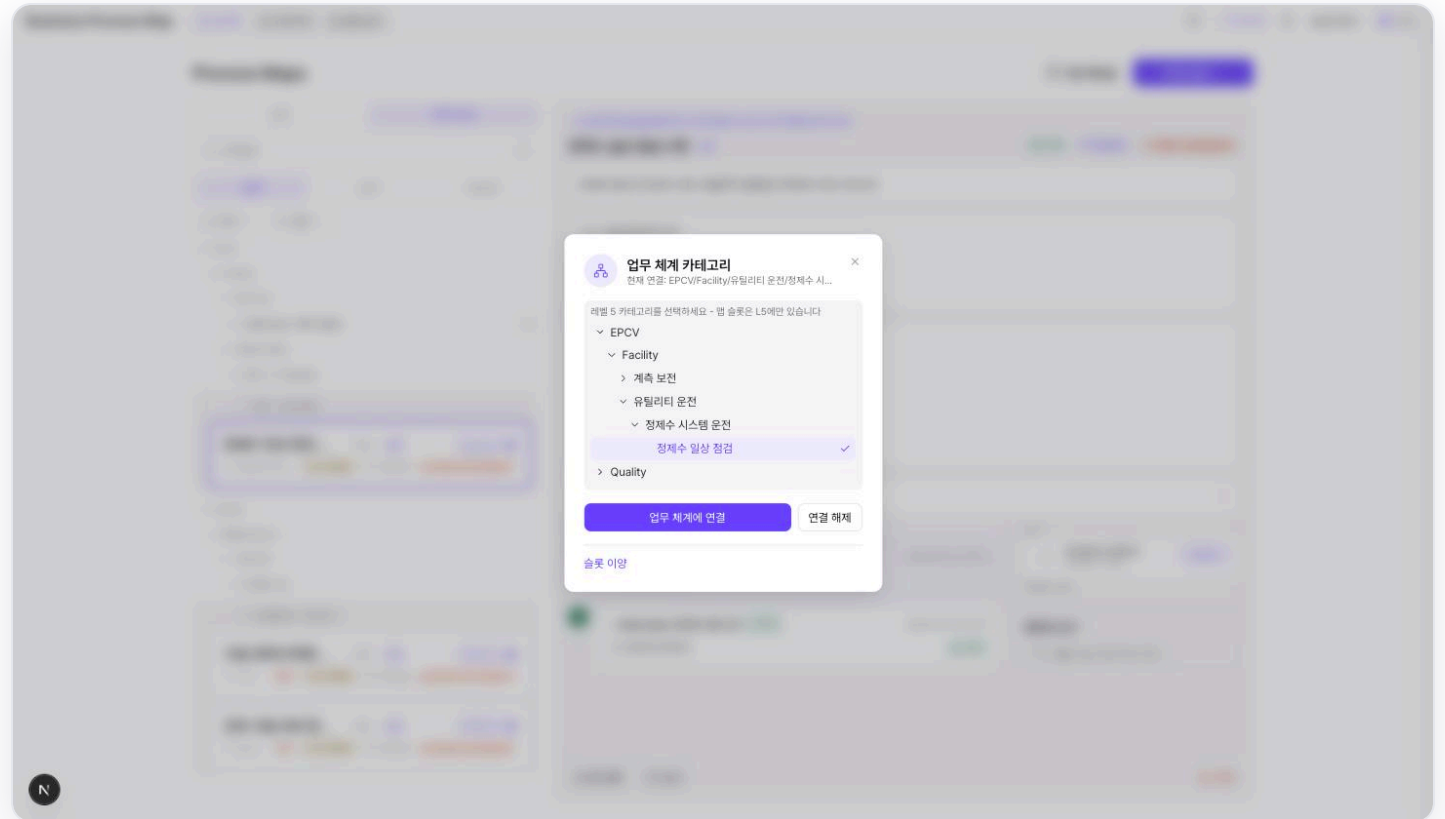
Personal dashboard & delayed navigation

- Order: Recently opened | Framework → My department | My documents → Needs approval | Recent changes; profile and tiles stay pinned on top.
- **My department** counts only your department's own maps — sub-department maps fold into a count line that opens a **dialog**; a dropdown widens the scope.
- My department · My documents have **Status | Version** tabs (unpublished · latest published · updating · recheck needed); click a legend to filter.
- A few rows per section, the rest behind "N more". A map in several sections gets a **left accent bar** on its other rows after a half-second hover.
- Delayed one-click navigation — tiles, Open buttons and row clicks run after a 0.6 s countdown; a translucent layer shows the ring and destination — click it to cancel.



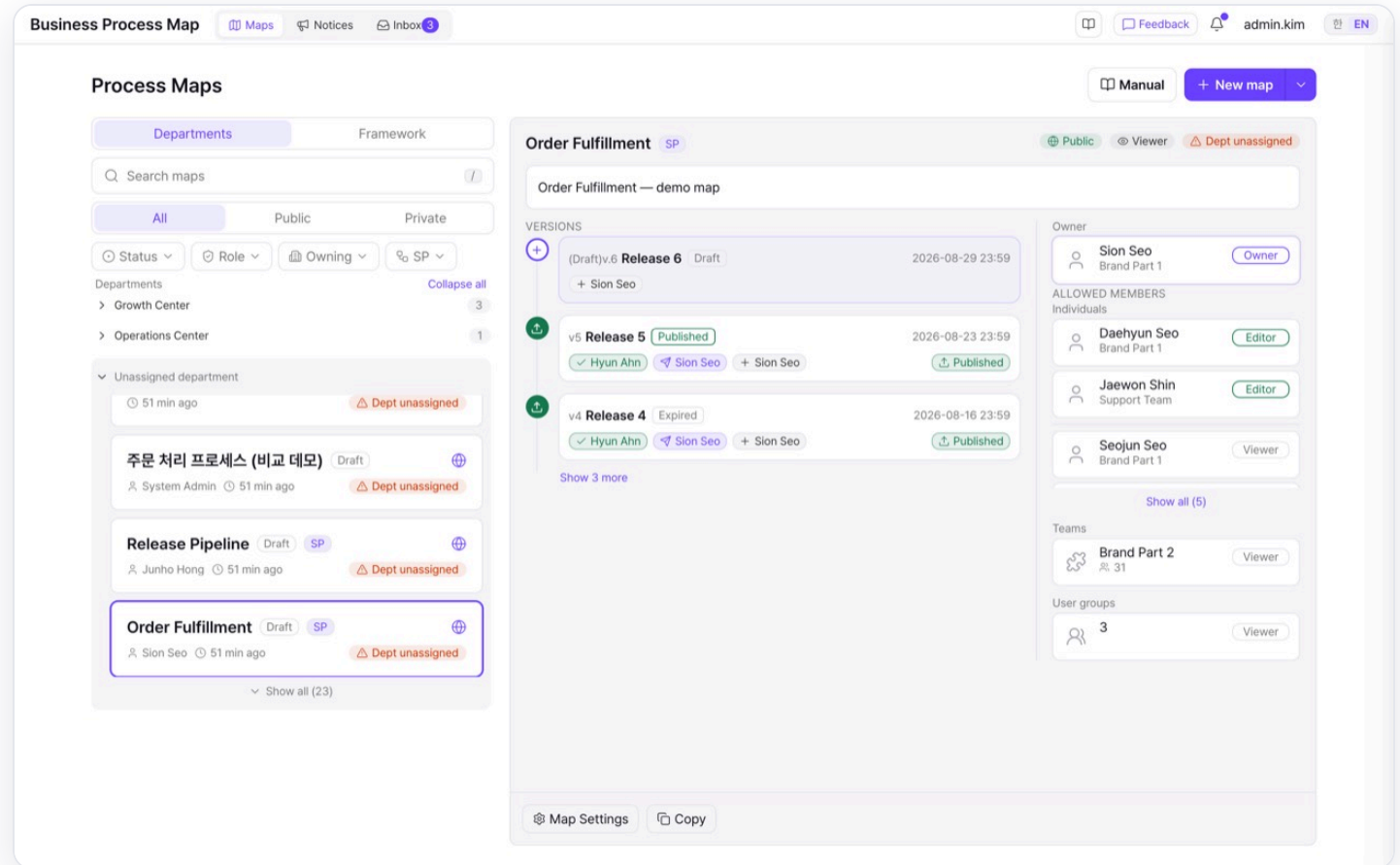
The Framework pill — assign, move, hand off

- The card's **Framework** pill opens the assign dialog — **Add to framework** (assign or move to another L5), **Unassign**, **Transfer slot**. L5 only.
- That L5's admin (or a sysadmin) applies it right away via **Apply now** in the confirm dialog (toast); anyone else files an **approval request** decided by the L5 admins (an L5 without admins falls to sysadmin).
- A move needs **both L5s' admins** (one person covering both counts once). A **progress (n/m)** banner shows while pending; the requester, the current owner or a sysadmin can withdraw.
- **Deleting** or **copy-and-retiring** a slotted map takes the same path — optional successor on delete; **Copy inherits the slot / Free the slot** on copy.
- Click a category row for the **category summary card** — children, subtree L5s and maps, admins, canvas state, **subtree confirmation** (🕒).



People cards & org info

- Hover a name briefly (0.7 s) or click for the **person card** — KO/EN names, title, org path, Open messenger.
- Right-click a name — **Send message** · **Info**.
- Click a department card for the **org info modal** — members, sub-organizations, and the path upward.
- Names follow your language setting — Korean UI puts Korean names first.



Creating a map

- **New map** — name, description, and visibility.
- **Owning department is required** — an org-tree picker; its members always hold Editor.
- **At least one approver is required** — a map cannot be created without one.
- Optionally add initial collaborators as Viewer/Editor — the picker sorts by **org proximity to you**.
- New maps come with **Start and End nodes pre-seeded**.
- **Create from CSV** — drop a CSV to create a pre-filled map instantly.

The image shows a 'New map' dialog box overlaid on a blurred background of a process map. The dialog box is titled 'New map' and has a close button (X) in the top right corner. It contains the following fields and options:

- Name:** A text input field containing 'Purchasing Approval'.
- Description:** A text input field with the placeholder text 'What is this process map about? (optional)'.
- Owning department (required):** A search input field with the placeholder text 'Search by name or initial consonant...'.
- Visibility:** Two radio buttons: 'Public' (selected) and 'Private'.
- Initial collaborators:** A section with a search input field and the text 'Search below to add editors or viewers.'.
- Required approvers (≥1):** A section with a search input field and the text 'Search below to add approvers.'.
- Footer:** A red text label 'Select the owning department.' on the left, and 'Cancel' and 'Create' buttons on the right.

Copying & replacing the original

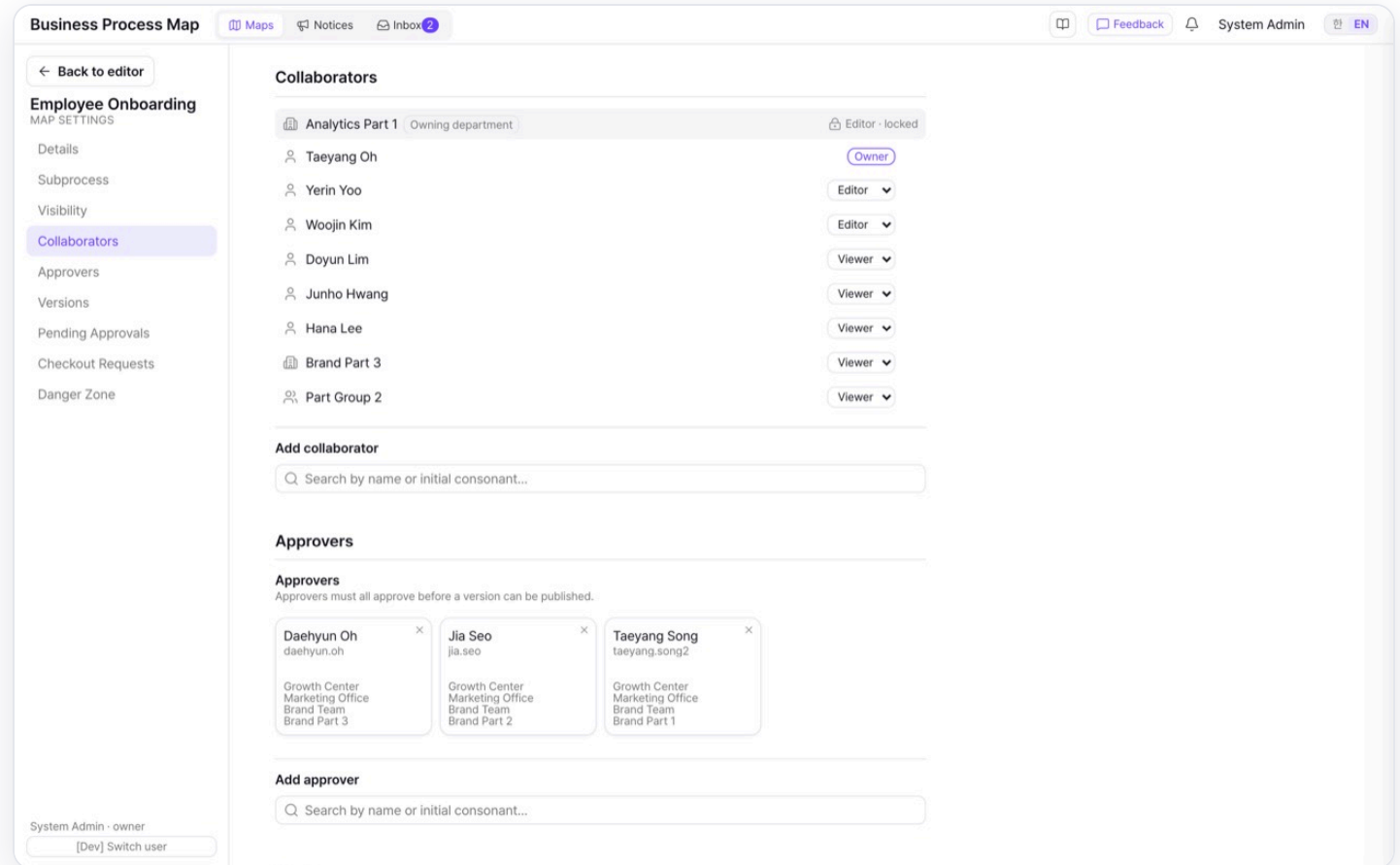
- Copy needs at least one published version — pick which one; the copy opens as the new map's draft. The original's owner is notified.
- Check "Move the original to the trash" — the copy replaces the original: it keeps the name; the original is tagged "(Pending deletion)" and trashed (7 days).
- The original's collaborators and approvers are pre-loaded to carry over after review; they are notified.
- A designated subprocess original asks you to confirm against its referencing processes.
- A **slotted** original goes through approval — choose **Copy inherits the slot** or **Free the slot**; linked L5 nodes stay stale until a successor is set.
- Plain deletion keeps the map in the trash for 7 days — restore from Settings → Scheduled deletion.

The screenshot shows a 'Copy map' dialog box with the following fields and options:

- Name:** A text input field containing 'Order Fulfillment'. Below it, a note states: 'The original moves to the trash, so the copy keeps the current map name.'
- Version to copy:** A dropdown menu showing 'v5 · Release 5 · published'. Below it, a note states: 'The copy opens as a new draft of the new map.'
- Owning department (required):** A search input field with the placeholder text 'Search by name or initial consonant...'.
- Owner:** A selection field showing 'System Admin admin.sys' with a 'You' button. Below it, a note states: 'The owner of the original map will be notified of this copy.'
- Visibility:** Two buttons, 'Public' and 'Private', with 'Private' selected.
- Move the original map to the trash after copying:** A checked checkbox with a trash icon.
- Details:** A list of informational items:
 - The original is renamed with a "(Pending deletion)" tag
 - Kept in the trash for 7 days, then permanently removed
 - Approvers and editor+ collaborators will be notified
 - Collaborators and approvers are carried over below — adjust before copying
- Warning:** A yellow box with a triangle icon stating: 'This map is a designated subprocess. Moving it to the trash removes it from the processes referencing it.'
- Footer:** A red text prompt 'Select the owning department.' and two buttons: 'Cancel' and 'Copy'.

Map Settings — Collaborators

- Add individuals, teams, or user groups as **Viewer** / **Editor** — the default list sorts by org proximity.
- Changes are **staged** (To add / Change / Remove pills), applied with **Save changes**, dropped with **Discard**.
- Each pill forecasts the outcome — ⚡ instant, or ⌚ **needs approval** (removing/demoting an Editor).
- Pending requests show an **Approval pending** tag; hover your own to **Withdraw** it.
- Right after saving, **Undo** reverts that one save.



Map Settings — name, visibility & more

- Renaming applies instantly only for the owner/admin (an editor files a request). Changing the **owning department** is owner/sysadmin-only — derived Editor access follows the new department.
- Interview-imported maps get a **Conditions & GMP** card on the Details tab — settle the GMP class, start/end conditions and times next to the source text (kept across re-imports).
- Visibility changes need approval — impact preview plus the deciding approvers; withdraw while pending. The Approvers tab locks during approval.
- Subprocess designation — needs a published version. The dialog has three sections — **Attributes / Metric tiles / Detail tiles** — each tile opens an in-place popover (⑤).
- Danger Zone — transfer ownership, delete map (Owner only).

The screenshot shows the 'Business Process Map' settings interface. The left sidebar contains a 'MAP SETTINGS' section with tabs for 'Details', 'Subprocess', 'Visibility' (selected), 'Collaborators', 'Approvers', 'Versions', 'Pending Approvals', 'Checkout Requests', and 'Danger Zone'. The main content area is divided into two sections: 'Visibility' and 'Collaborators'.

Visibility Section:

- Header: 'Visibility' with a sub-note 'Changing visibility requires approval.'
- Buttons: 'Private · current' (selected) and 'Public'.

Collaborators Section:

- Header: 'Collaborators'.
- Table of collaborators with columns for name, role, and department.

Analytics Part 1	Owning department	Editor · locked
Taeyang Oh		Owner
Yerin Yoo		Editor
Woojin Kim		Editor
Doyun Lim		Viewer
Junho Hwang		Viewer
Hana Lee		Viewer
Brand Part 3		Viewer
Part Group 2		Viewer

Add collaborator

Search by name or initial consonant...

Approvers

Approvers must all approve before a version can be published.

Daehyun Oh	Jia Seo	Taeyang Song
daehyun.oh	jia.seo	taeyang.song2
Growth Center Marketing Office Brand Team	Growth Center Marketing Office Brand Team	Growth Center Marketing Office Brand Team

System Admin · owner
[Dev] Switch user

03

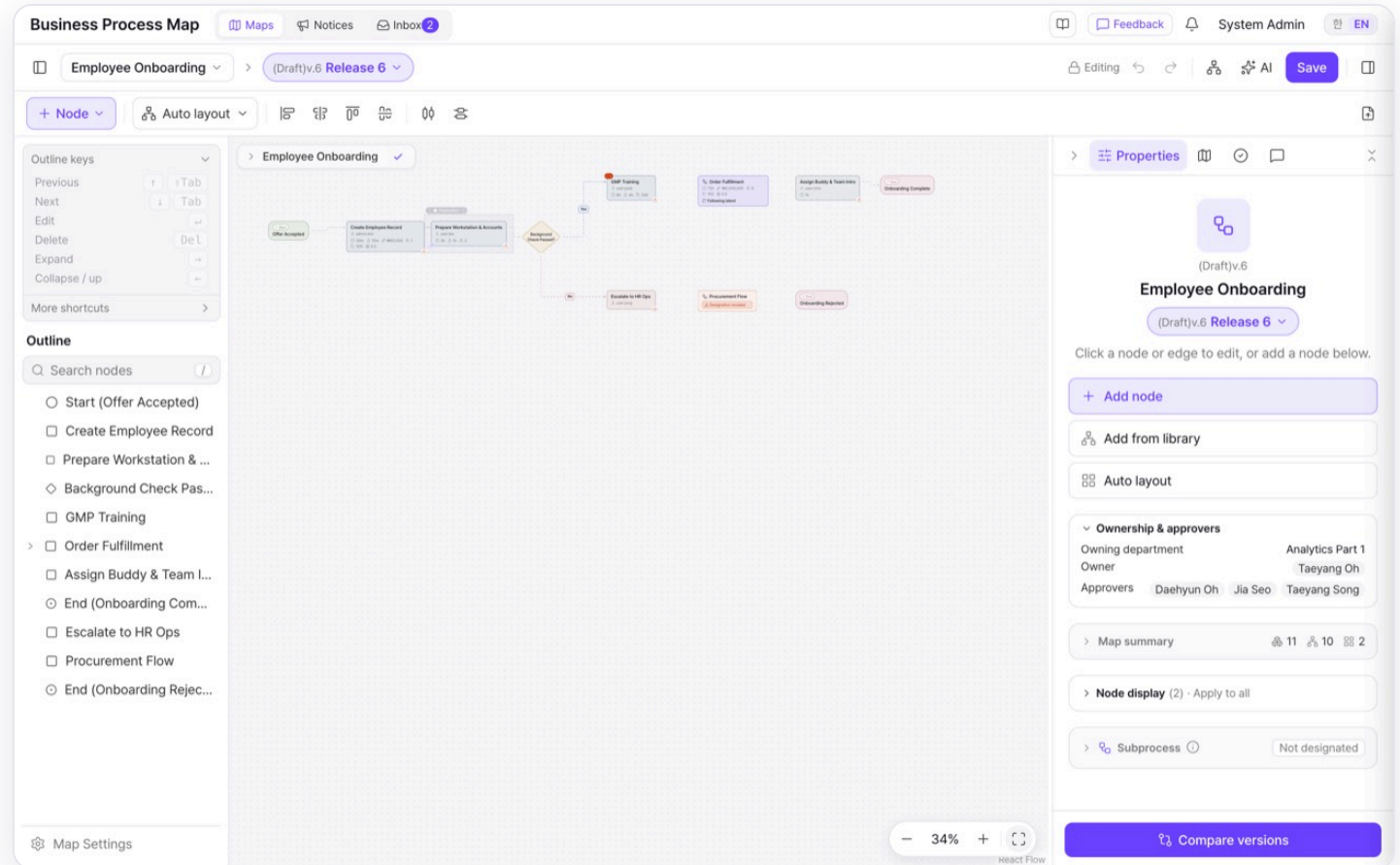
Editing — The Basics

The editor, canvas navigation, creating and connecting nodes, tidying and saving.

③ Editing Basics

The editor at a glance

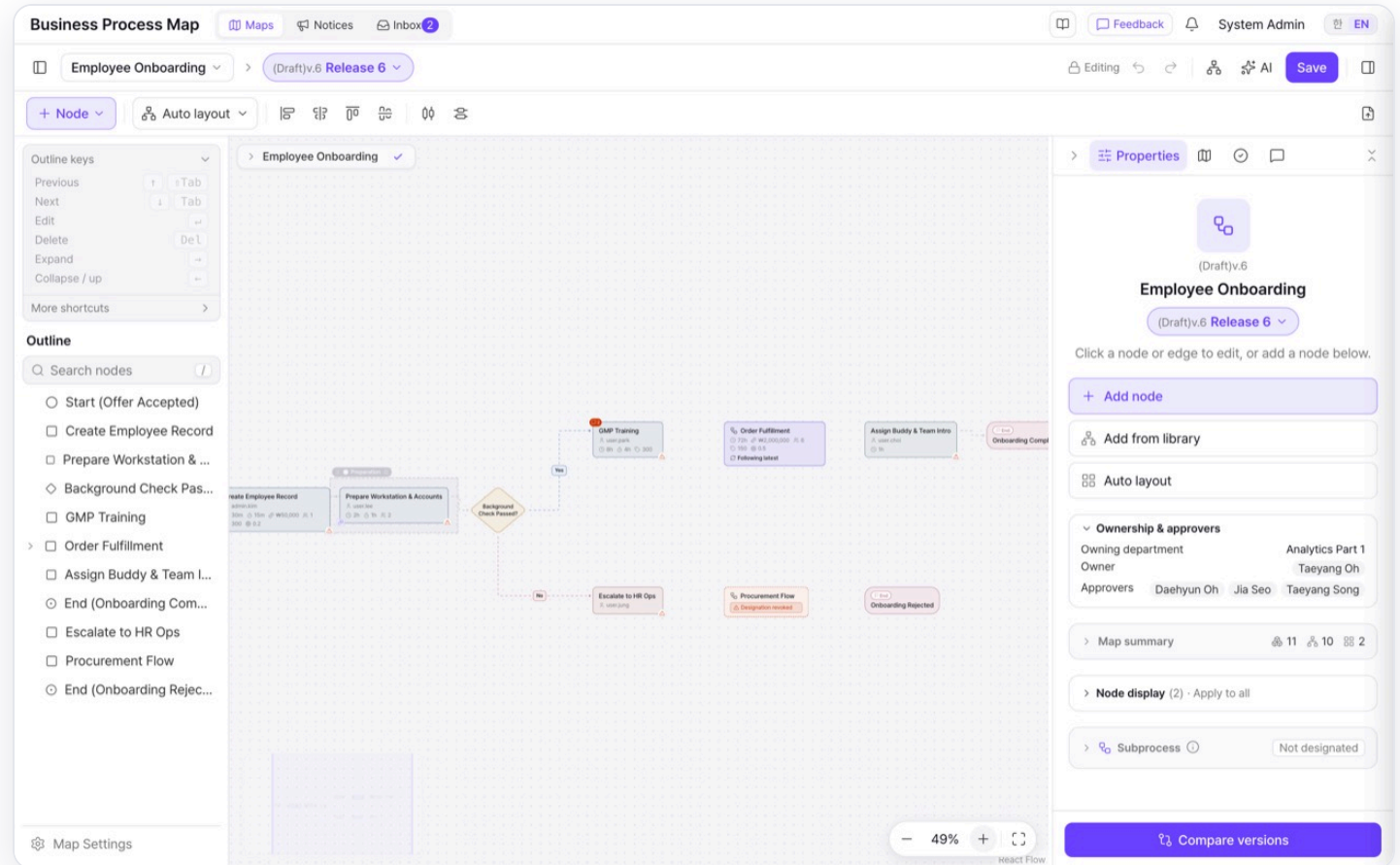
- Top bar — map menu, version selector, undo/redo, library, AI, Save.
- Left — the outline tree and shortcut card. Right — the inspector (Properties / Map / Approval / Activity). Node display fields toggle in the Map tab's **Node display** or the **eye icon** card next to the zoom pill (both show the same values).
- Framework-registered maps show the **Framework chip top-left** with the classification path (title and save checklist sit top-right) — expand it to jump to sibling maps, or **Search other frameworks** at the bottom.
- A brand-new map with only Start/End shows the **"Start with the AI consultant"** balloon — Start opens the interview (⑧).
- The rightmost icon on the inspector tab bar collapses/expands every section of the tab.



③ Editing Basics

Moving around the canvas

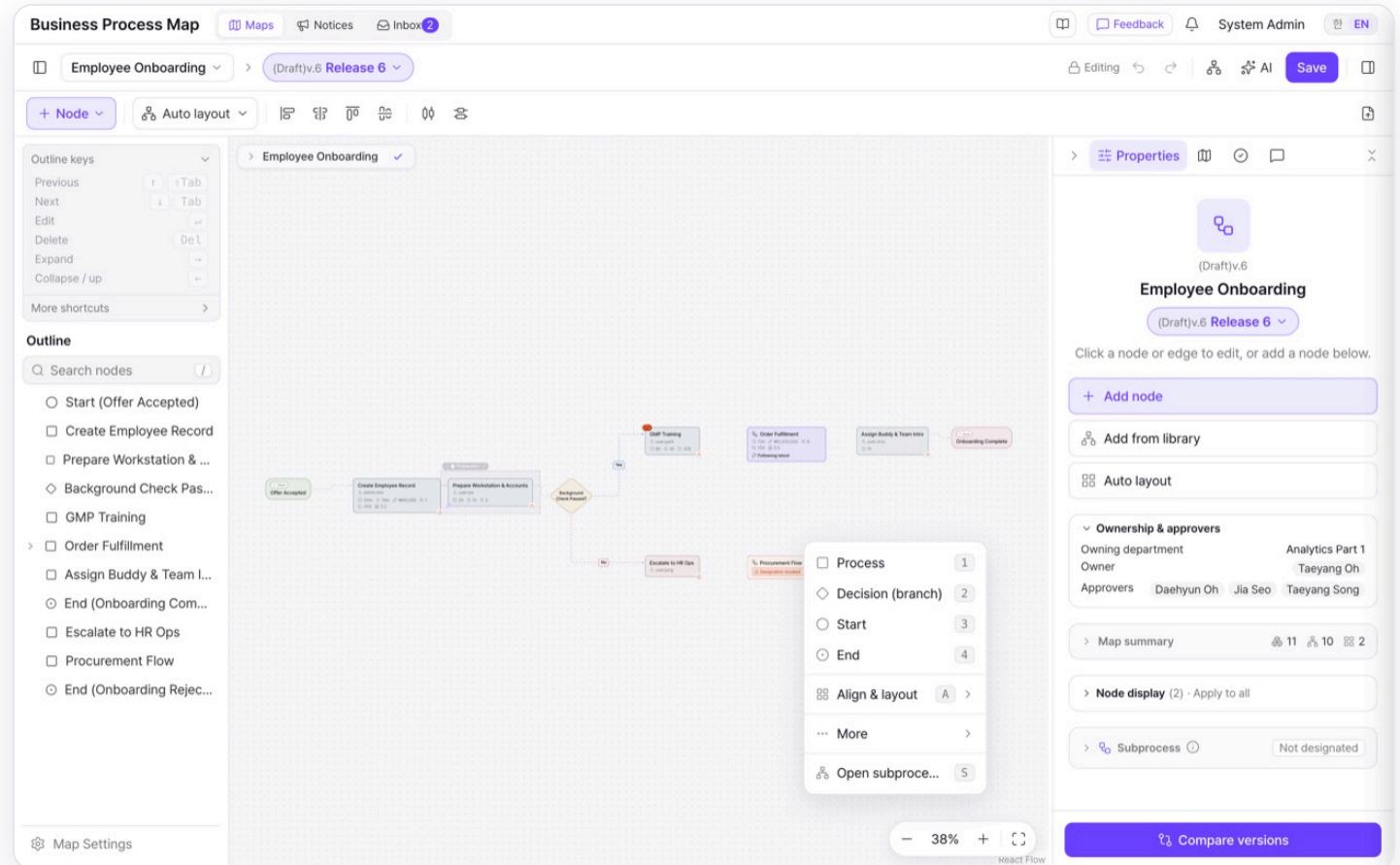
- Wheel = vertical pan, **Shift** +wheel = horizontal, **Ctrl** +wheel = zoom.
- Hold **Space** and drag to pan; drag empty canvas to **box-select**.
- Right-click opens the context menu (node, edge, or canvas).
- When a node grows, nodes below **shift on screen** so nothing overlaps (saved coordinates unchanged).



③ Editing Basics

Creating nodes

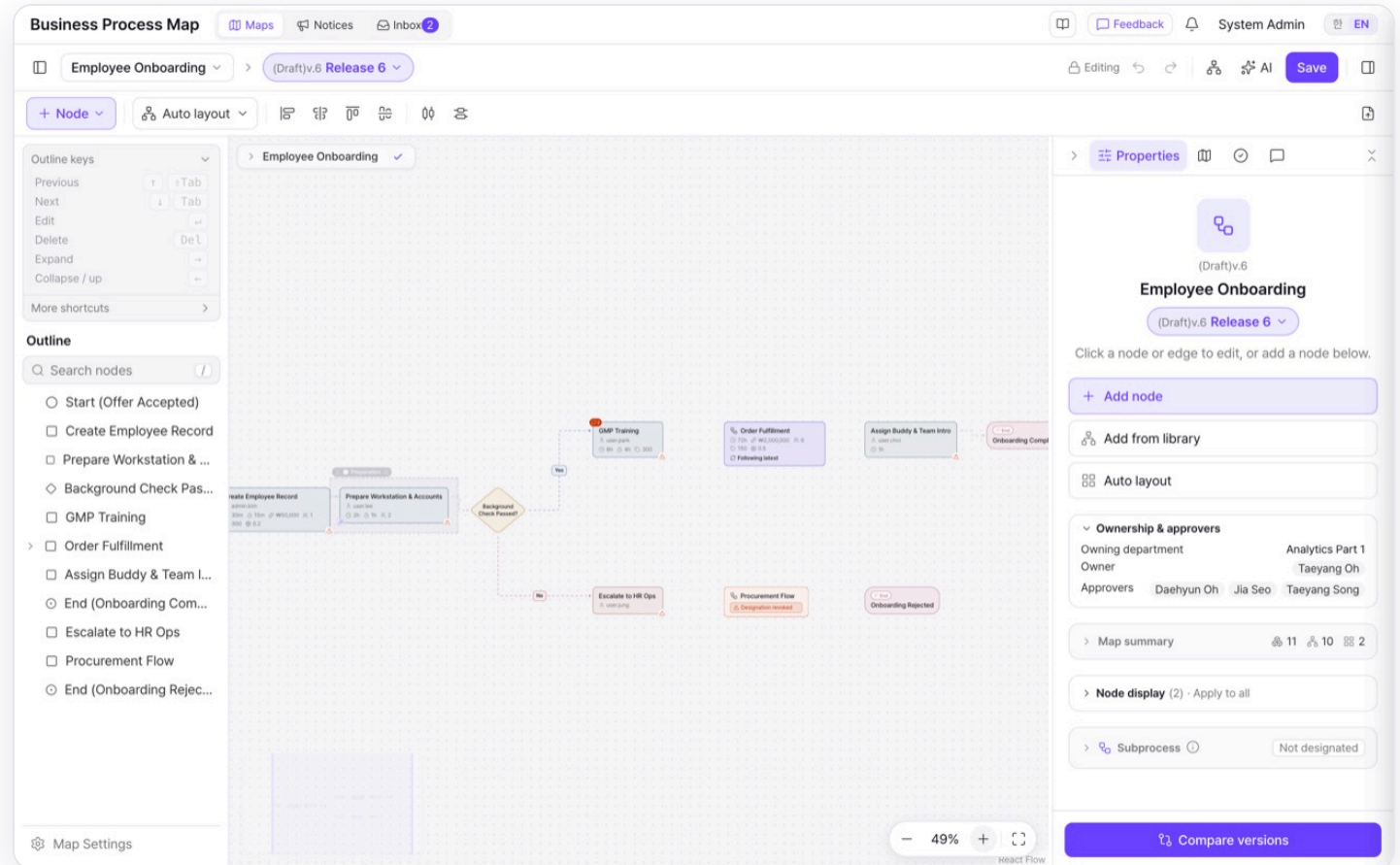
- **Right-click** the canvas, or use **Add node** in the inspector.
- **Start** (pill, only one) · **Process** (rectangle) · **Decision** (diamond) · **End** (one primary end) · **Subprocess**.
- Double-click or **F2** renames in place — **Enter** commits, **Shift+Enter** breaks the line.
- Copy with **Ctrl+C/V** (across tabs and maps), duplicate with **Ctrl+drag**; duplicate names get a **(2)** suffix.



③ Editing Basics

Connecting nodes

- Drag from a handle — **dropping on a node's body** connects to the default handle (with a snap preview).
- Plain nodes have **one output** — branch with a **Decision** node; its edges carry Yes/No labels.
- Edit edge labels via right-click or **F2**; line breaks supported.
- **Line style is per edge** — curved/stepped/straight from the inspector or right-click; change all at once in the Map tab's **Edge style**.
- Drag a node near another for **drop zones** (before/after/swap/group) to insert in one motion.



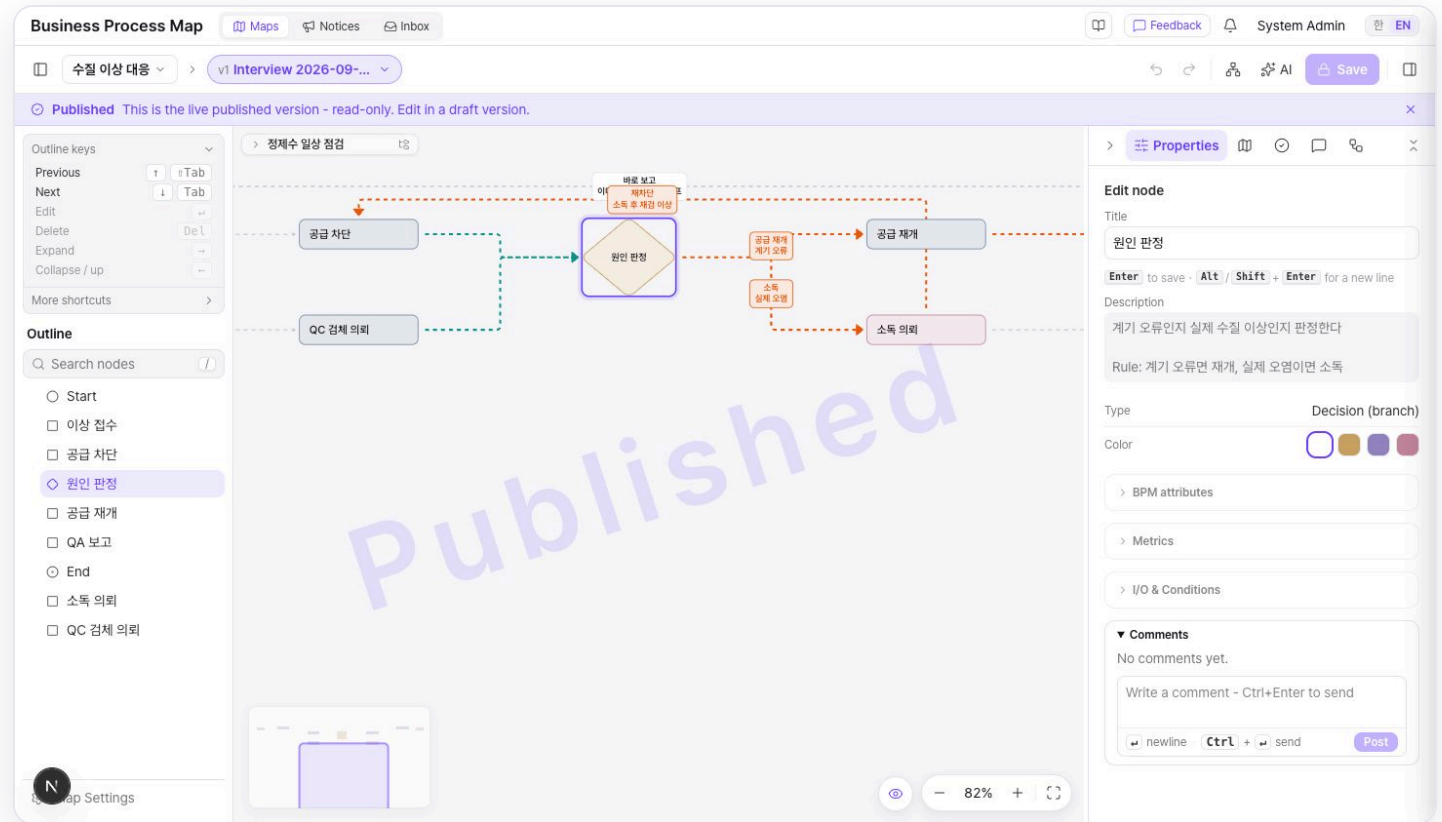
Tidying — layout, groups, bulk edit

- **Auto layout** — horizontal `Shift+L`, vertical `Shift+K`; the main path snaps to one line.
- `Shift` +drag = axis-locked move; `Alt+W/C/T/X` align; `Alt+R/V` distribute.
- Select 2+ nodes and press `Ctrl+G` for a **group** — double-click the title to rename, drag it to move.
- **Group bulk edit** — pick a field under Attributes / Metrics / I/O & Conditions and set or clear it across all members.
- Append / Replace / Skip conflict handling with a before/after summary.

③ Editing Basics

Search & following the flow

- **/** — node search (Korean chosung supported); matches jump and highlight.
- Click an **outline** entry to focus that node.
- **Tab / Shift+Tab** — next/previous node along the flow (including expanded subprocess children).
- **]** / **[** — grow or shrink the flow highlight from the selected node (incoming teal, outgoing orange). A selected connector turns violet, and the **label pill** takes the same color as its connector.
- Per-node **comments** — **Ctrl+Enter** sends; read-only users can comment too.



Saving & validation

- The editor **autosaves** ~2 s after you stop; the Save button shows Saving.../Saved/Failed.
- **Save checklist** — one Start, one primary End, unique End names, no multi-output plain nodes.
- Editing needs a draft (#Draft/#Rejected) plus **your checkout** — otherwise the canvas locks read-only.
- While a downgrade of your permission is pending, checkout and submission are temporarily refused.

Shortcuts at a glance

- **Editing** — `Ctrl+C` / `Ctrl+V` copy/paste (across tabs & maps), `Ctrl` +drag duplicate, `Ctrl+Z` / `Ctrl+Shift+Z` (or `Ctrl+Y`), `Delete`, `F2` rename/label, `Shift+Enter` line break.
- **Canvas** — hold `Space` to pan, `Ctrl` +wheel to zoom, `/` node search (chosung), `S` subprocess list, `Esc` cancel · close dialogs/popovers.
- **Tidying** — `Shift+L` / `Shift+K` auto layout (horizontal/vertical), `Ctrl+G` group, `Alt+W/C/T/X` align left/center/top/middle, `Alt+R/V` distribute, `Shift` +drag axis lock.
- **Navigation** — `Tab` / `Shift+Tab` next/previous along the flow, `] / [` grow/shrink the highlight, `Alt+←` / `Alt+→` collapse outline/inspector, `Ctrl+Shift+E` PNG, `Ctrl+Enter` send.
- **More shortcuts** at the bottom of the Outline keys card shows the same list in context.

04

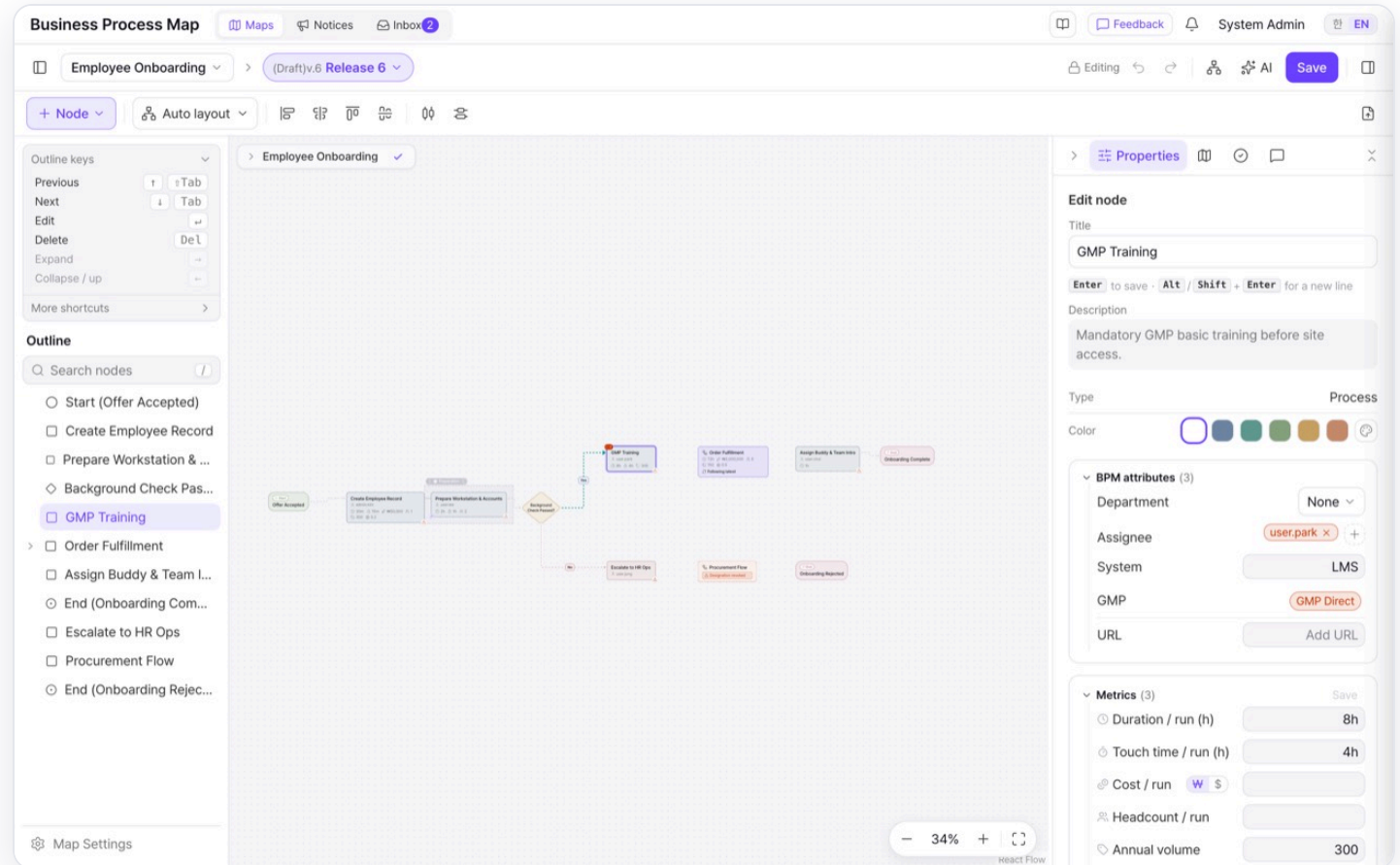
Node Data

Attributes, per-run metrics, GMP classification, I/O & conditions, and links.

④ Node Data

Node attributes

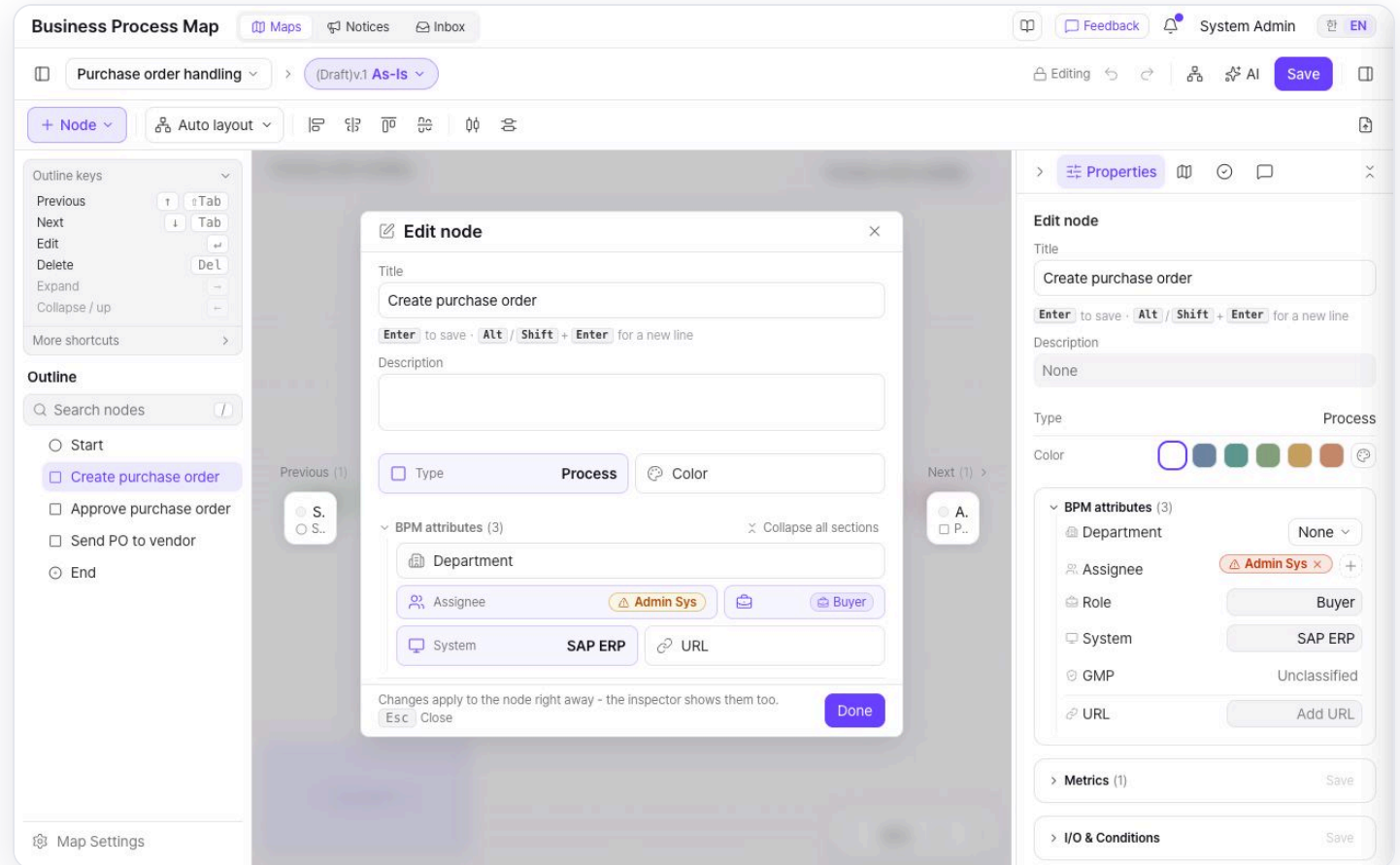
- Double-click the title or F2 renames in place — double-clicking the body or right-click **Edit info** opens the **edit dialog** (next slide).
- The description is read-only in the inspector — the hover icon or a double-click opens the dialog on it. Enter confirms, Shift+Enter breaks a line.
- BPM attributes — assignee (org directory), **Role**, department (auto), system. Color via presets/hex.
- **Role** — a role name, separate from the assignee; catalog autocomplete (alias → canonical), free text kept. Nodes show it at rest, the assignee on hover.
- **System** is matched to the system list — alias → canonical; unknown values become **Other** + source note.
- GMP — Direct/Indirect/Non-GMP pill, color locks to the class (undo). URL badge; subprocess nodes inherit the linked map read-only.



④ Node Data

The node edit dialog (tiles)

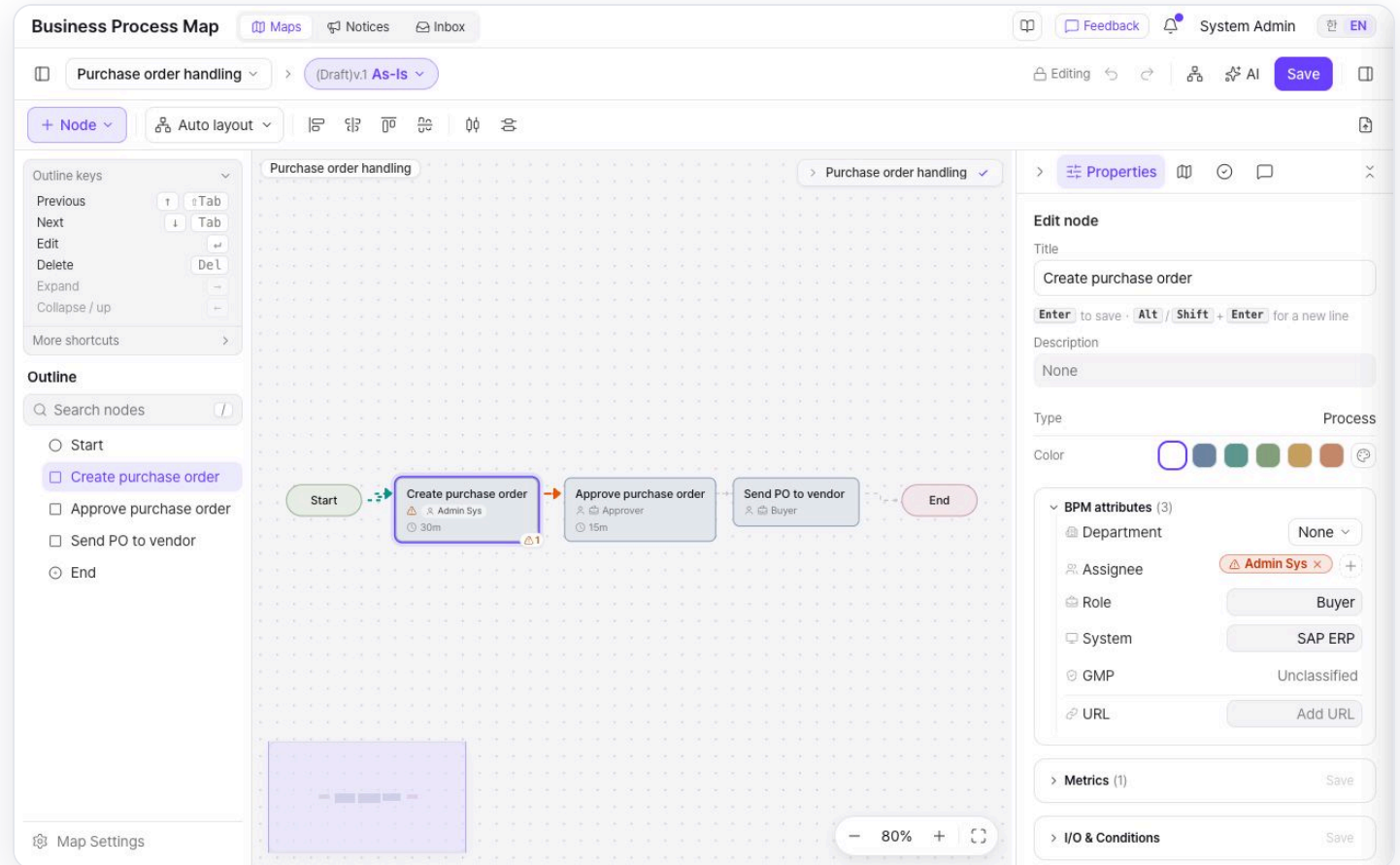
- Fields sit as **tiles** below title and description — type & color, **Attributes** (department, assignee, system, link, GMP), the **seven metrics**, **I/O & conditions**.
- Click a tile → a **popover where you clicked**: Enter applies (Ctrl/⌘+Enter for multi-line), Esc cancels, outside click applies only if changed. Save ▾ — save / save & close / close without saving.
- Cost is one tile — **W / \$ tabs** pick the currency; the other clears on save. The department picker puts your own chain first. Values sync to the node and inspector instantly — the footer is just **Done** (Esc); undo is **Ctrl+Z**.
- Hovering the System tile reveals the **interview source note** icon. **Previous / next nodes** float as dock cards — click one to slide into editing it.
- Read-only contexts show only filled tiles. Subprocess nodes show inherited tiles and edit only **annual volume** and **FTE**.



④ Node Data

Roles & the system catalog

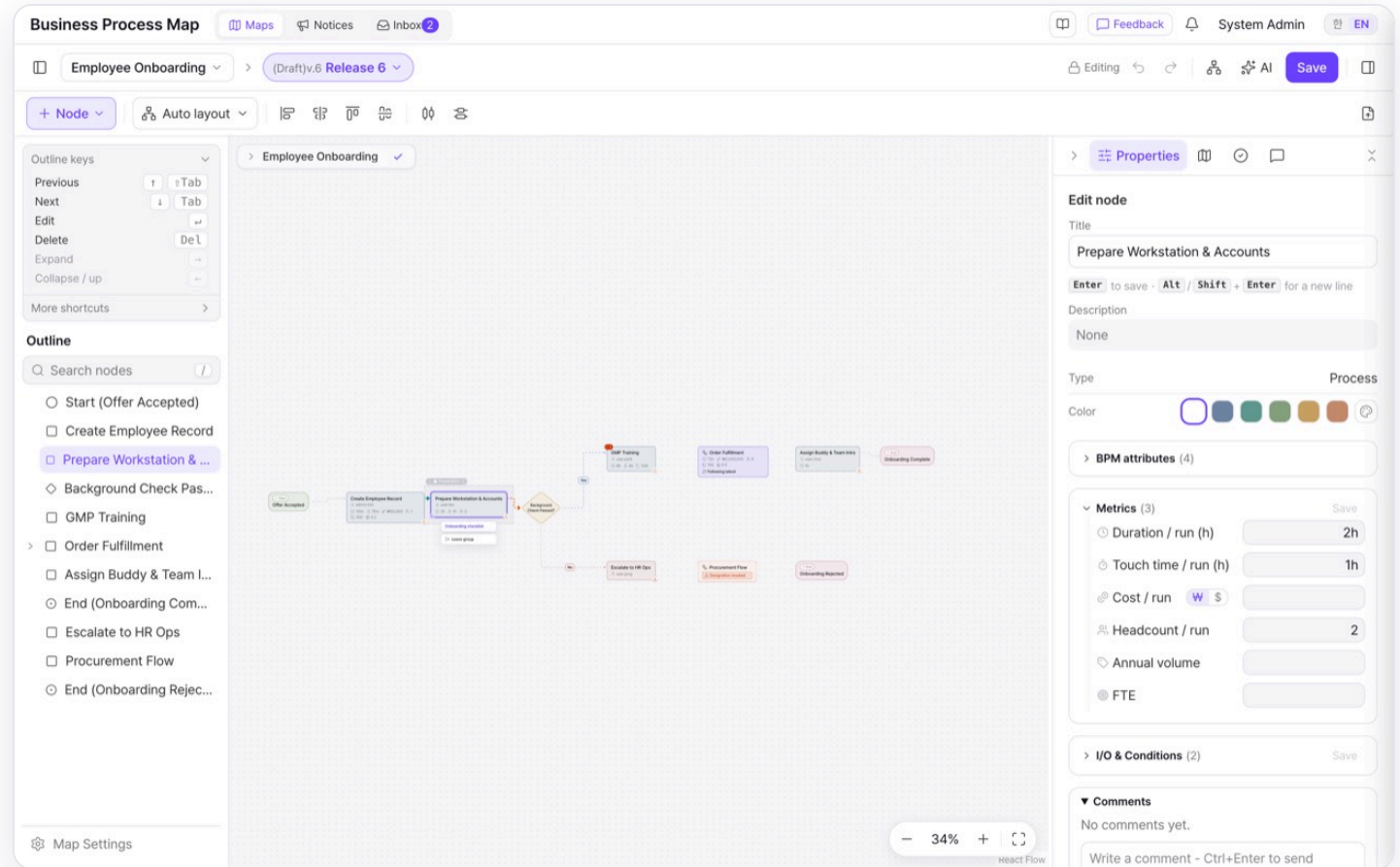
- Inspector **Role** row under Assignee — autocomplete from the admin role list (alias → canonical); values outside the list are saved as typed.
- Node edit and SP designation dialogs put **Assignee 2 : Role 1** on one tile row — a filled tile shows only the chip. Group bulk edit has a Role mode.
- A system outside the list is stored as **Other + source note** — with a different note already present, choose replace / append / cancel.
- Nodes show the role at rest and swap to the **assignee chip** ~1 s after hover/select; the role also shows in the library, peek, home detail and Excel.
- Departments or assignees missing from the org chart get **warning-colored** icons and a Stale refs badge — roles are not checked, so no warning.



④ Node Data

Per-run metrics (seven)

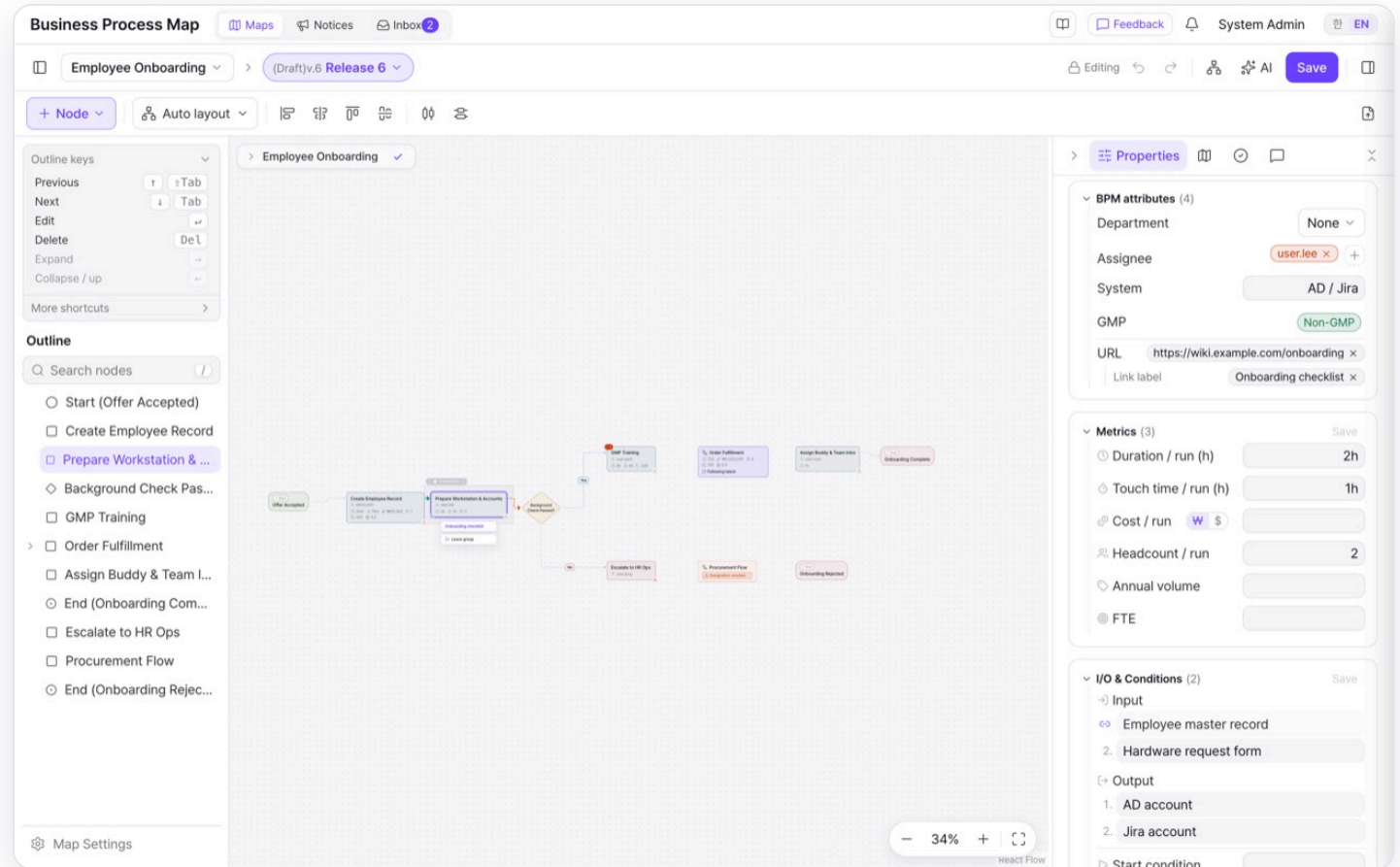
- Duration, touch time, cost (₩/\$), headcount, annual volume, FTE — **per single run**.
- **h.mm notation** — `1.30` = 1 h 30 m; `0.90` carries to 1h30m. Shows as typed while editing, `1h30m` elsewhere.
- **One currency only** — a ₩/\$ toggle; switching warns that the other value clears, with Undo.
- Metrics and I/O sections apply via their **section Save button** (lit when dirty).
- Subprocess nodes take only **annual volume & FTE** directly — the other five inherit from the linked map.



④ Node Data

Inputs, outputs & conditions

- Inputs/outputs hold one item per row laid out as **number / data-form pill / item name** — click the dotted form slot to set the data form (.xlsx/Excel/엑셀 all match).
- Import from node... pulls an upstream output / downstream input and stays Linked; Disconnect detaches a copy.
- Linked items carry a link icon with a direction tooltip; hovering lights up the peer node and path on canvas.
- Inputs take a Required/Optional flag — told apart by hover color (required = red tint) and tooltip.
- Start/end conditions are free text. On canvas they show below the selected node (choose items in Node display).



05

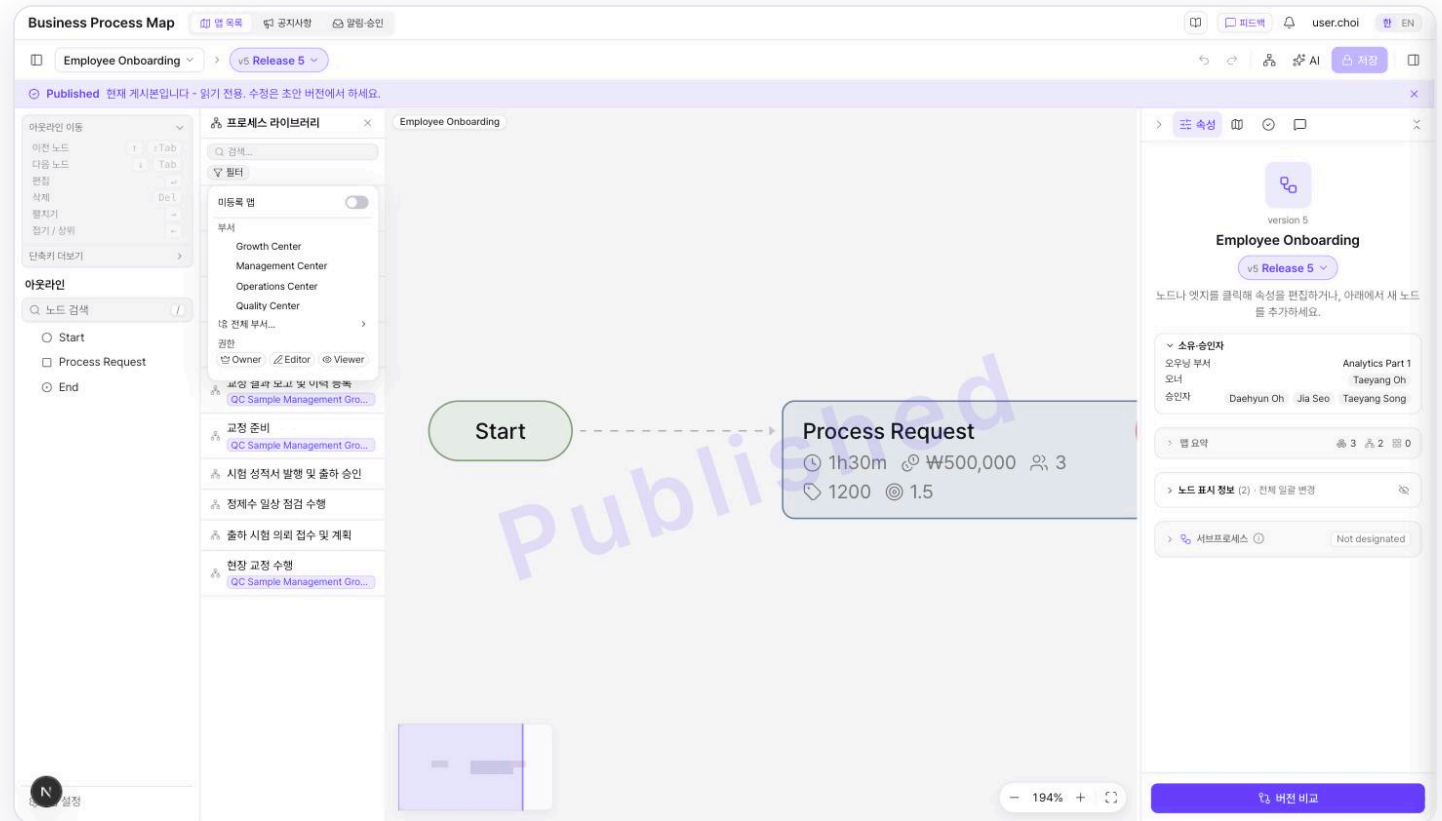
Subprocesses

Reference another map as a single step — preview it, expand it, manage registration.

⑤ Subprocesses

Adding as a link

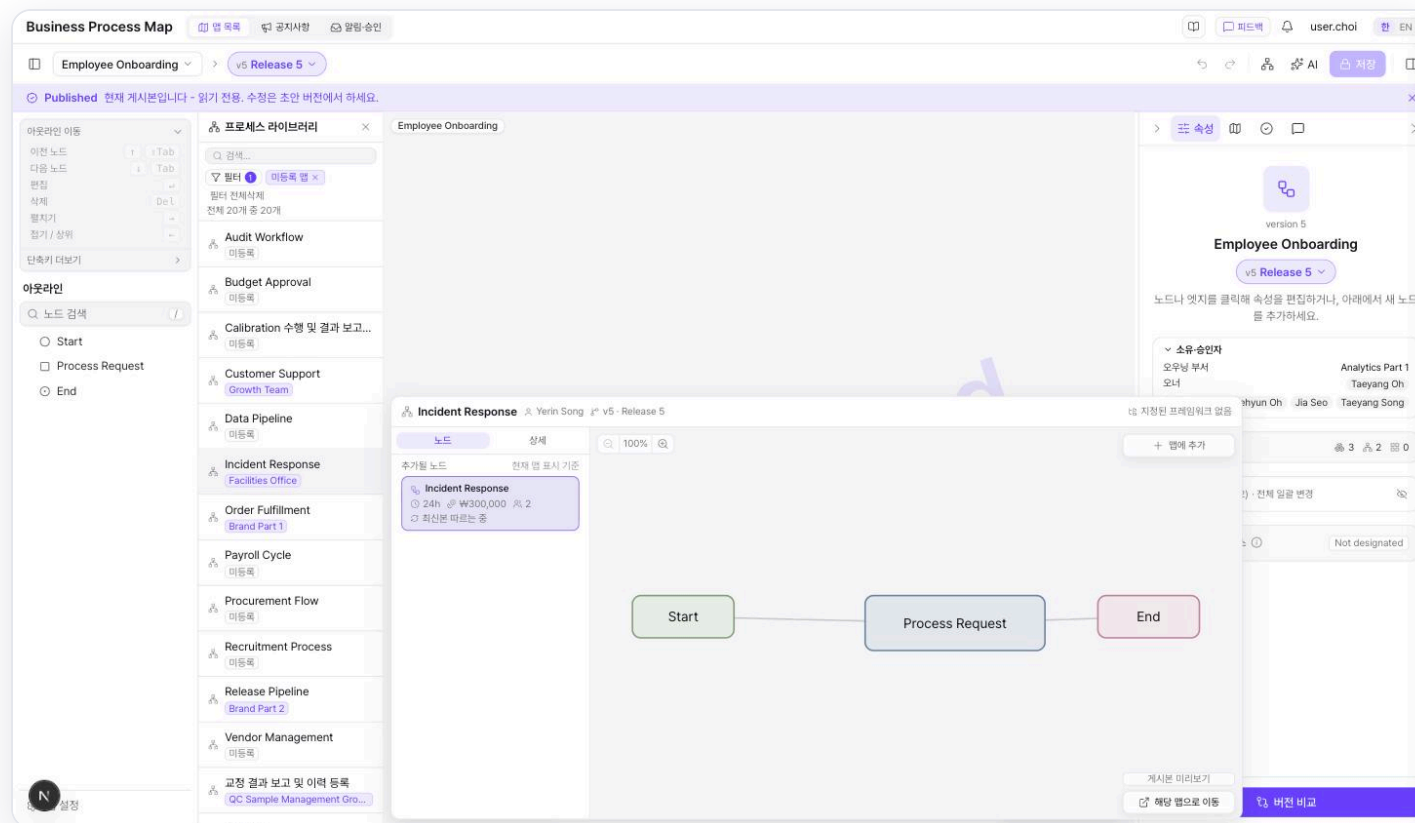
- A reference, not a copy — changes to the source map flow through.
- Drag from the library (S, chosung search) — designated maps by default. **Filter:** an **Unregistered maps** switch on top, **Department** rows (your chain; **All departments...** opens the org-tree flyout with search), **Role** pills.
- Active filters show as removable pills (remembered per browser). New links follow the latest published version; pinning is possible. Duplicate links are blocked.
- The node's context menu ends with **Open linked map** — confirmed before leaving the editor (unsaved changes are protected).
- Selecting the node shows the Subprocess tab — designation info · maps linking this one · designated parameters (single-column tiles, gray tiles for missing values).



⑤ Subprocesses

Preview peek — Node tab

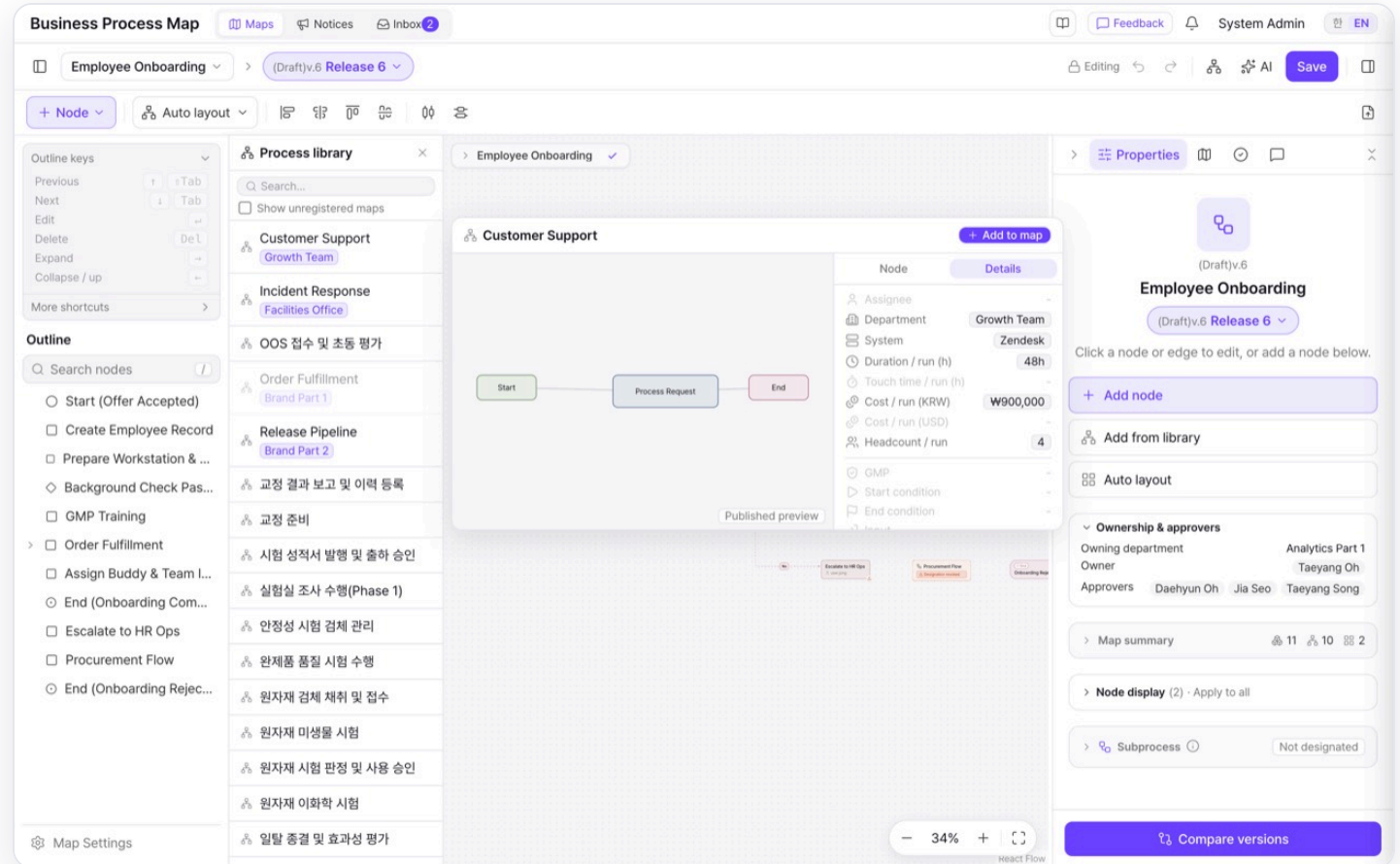
- Click a library row (or hover 2.5 s) for the published-version peek. Header: map name, owner and version on the left, the **framework path** on the right (opens the tree and "Search other frameworks").
- The flow preview **zooms with the wheel** (100–300%) and pans by drag; **Add to map** and **Open this map** float above it.
- Node tab — a live mock of the node you'd add: attributes, metric chips, GMP, conditions and the I/O checklist; hovering switches it to this map's display settings.
- You can also **drag the mock straight onto the canvas**. Maps you can't view show a lock notice but can still be added.
- Hovering a **department chip** in the list opens the org info card.



⑤ Subprocesses

Preview peek — Details tab

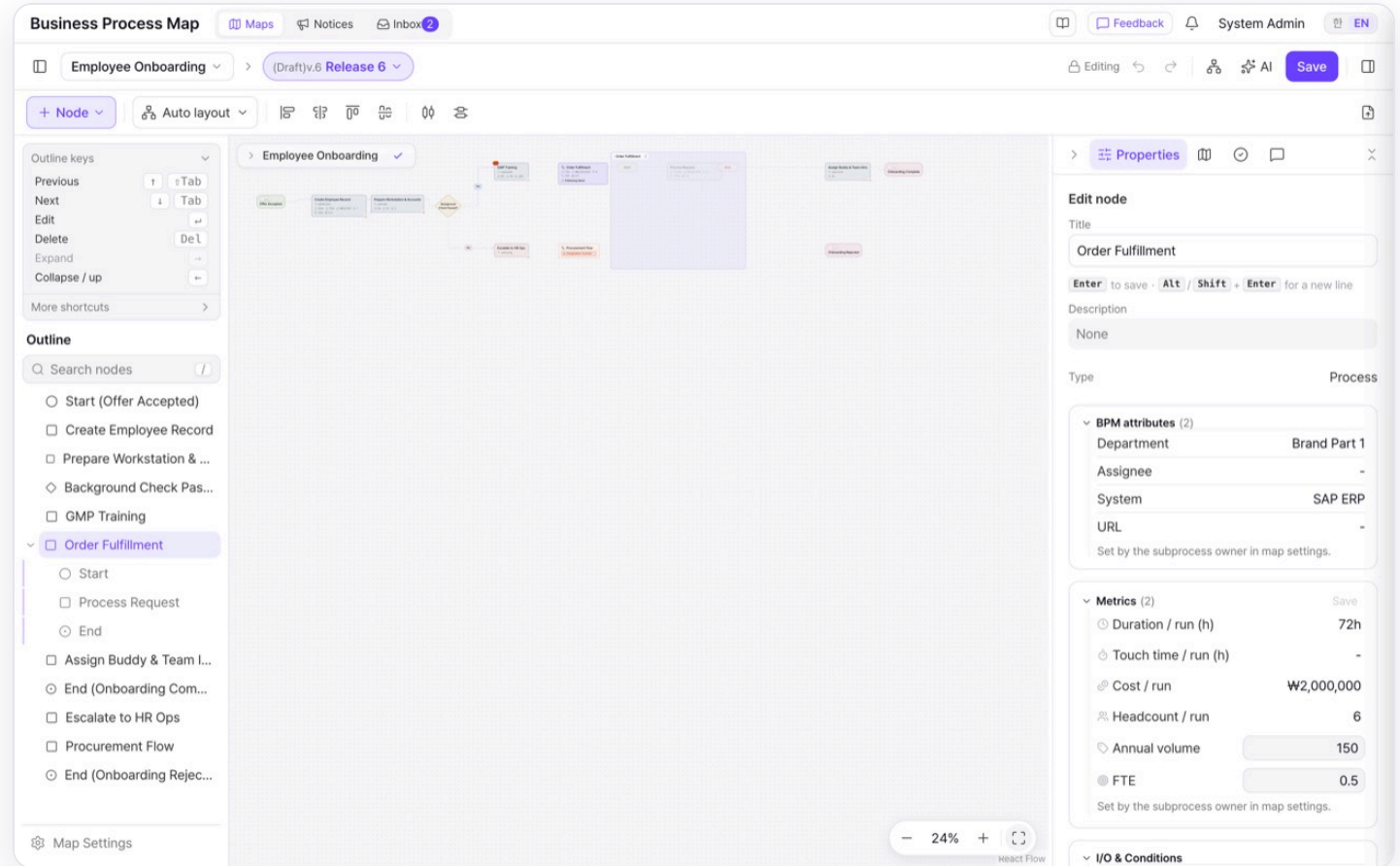
- Designated attributes (department, assignee, system) and **subprocess metrics**.
- Conditions, I/O, and GMP in full — empty values tone down to dashes.
- Meta — **published version (v-number), publish time, framework path**.



⑤ Subprocesses

Expand in place (inline)

- Right-click → **Expand subprocess** (also on the action bar under the selected node) — the published child embeds read-only in a tinted region on this canvas.
- The region header shows the map name with an open icon — framework maps add the full 5-level path label (click for the peek).
- Right-click inside — Open linked map (confirmation gate) · Collapse. Hovering highlights the region.
- Tab traversal includes expanded children; child GMP pills are read-only.
- Double-click opens the **node edit dialog** — there is no separate deep-view overlay; the child's content is read via expand only.



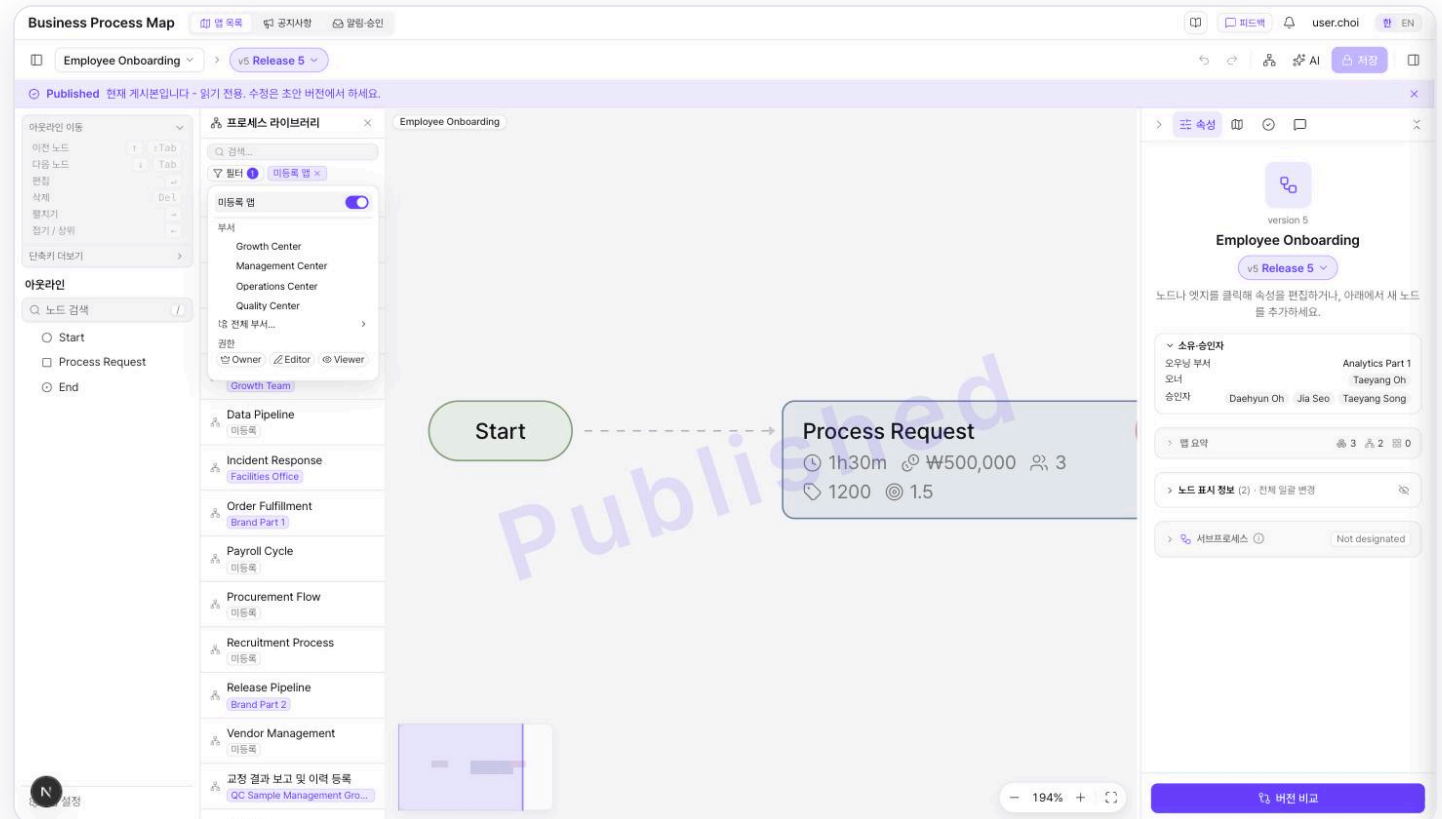
Node polish & opening the linked map

- **Node width** — hover for the right-edge grip and drag (180 px–120%); the width saves with the map.
- **Framework pill** — framework-registered links show a folder pill; click/hover 3 s for the path peek (0.4 s grace close) and "Search other frameworks".
- **Open linked map** — the last item of the context menu; disabled without a link or view access, and confirmed before leaving the editor.
- Without permission on the linked map the node shows **No access**. Double-click opens the node edit dialog (annual volume and FTE only).

⑤ Subprocesses

Linking unregistered maps

- Turn on the **Unregistered maps** switch at the top of the library's **Filter** popover — unregistered maps appear with a **"Not registered"** badge ("Clear filters" resets).
- Dropping one confirms the warning, then asks whether to send a registration request (Send request / Link only).
- The request lands in the target owner's Inbox — once designated, the warning clears and you're notified.
- Select the node for Request registration — with a status badge and withdraw afterwards.
- Create map "query" under the library search — makes a new map and links it here (the L5 canvas's L6 picker offers **Create placeholder** instead).



Managing designation (owner)

- Map Settings → Subprocess designation — requires a published version. Designated values render live on every linking node.
- Three sections — **Attributes** (department required, assignee, system, link), **Metric tiles** (one currency via ₩/\$ tabs, Σ suggestions, source notes), **Detail tiles** (I/O with per-item data forms).
- Click a tile → in-place popover (Enter saves, Esc cancels). **Annual volume** and FTE are **reference values** — each linking node keeps its own.
- No separate designation description — the **map description** is used in the library and on linking nodes.
- The card's Linked from accordion lists linking maps; the inspector's Subprocess tab (published version) opens the dialog via **Edit**.

The screenshot shows the 'Business Process Map' interface for 'Employee Onboarding'. The left sidebar contains a 'MAP SETTINGS' section with tabs for 'Details', 'Subprocess' (selected), 'Visibility', 'Collaborators', 'Approvers', 'Versions', 'Pending Approvals', 'Checkout Requests', and 'Danger Zone'. The main content area is titled 'Subprocess' and includes a description: 'Designate this map as a company subprocess so other maps can link it from the library.' Below this is a 'Designate as subprocess' button. The 'Visibility' section shows 'Private · current' as the selected option, with a 'Public' option available. The 'Collaborators' section lists various roles and their assigned users: 'Analytics Part 1' (Owning department, Editor - locked), 'Taeyang Oh' (Owner), 'Yerin Yoo' (Editor), 'Woojin Kim' (Editor), 'Doyun Lim' (Viewer), 'Junho Hwang' (Viewer), 'Hana Lee' (Viewer), 'Brand Part 3' (Viewer), and 'Part Group 2' (Viewer). At the bottom, there is an 'Add collaborator' section with a search bar labeled 'Search by name or initial consonant...'. The footer shows 'System Admin · owner' and a '[Dev] Switch user' button.

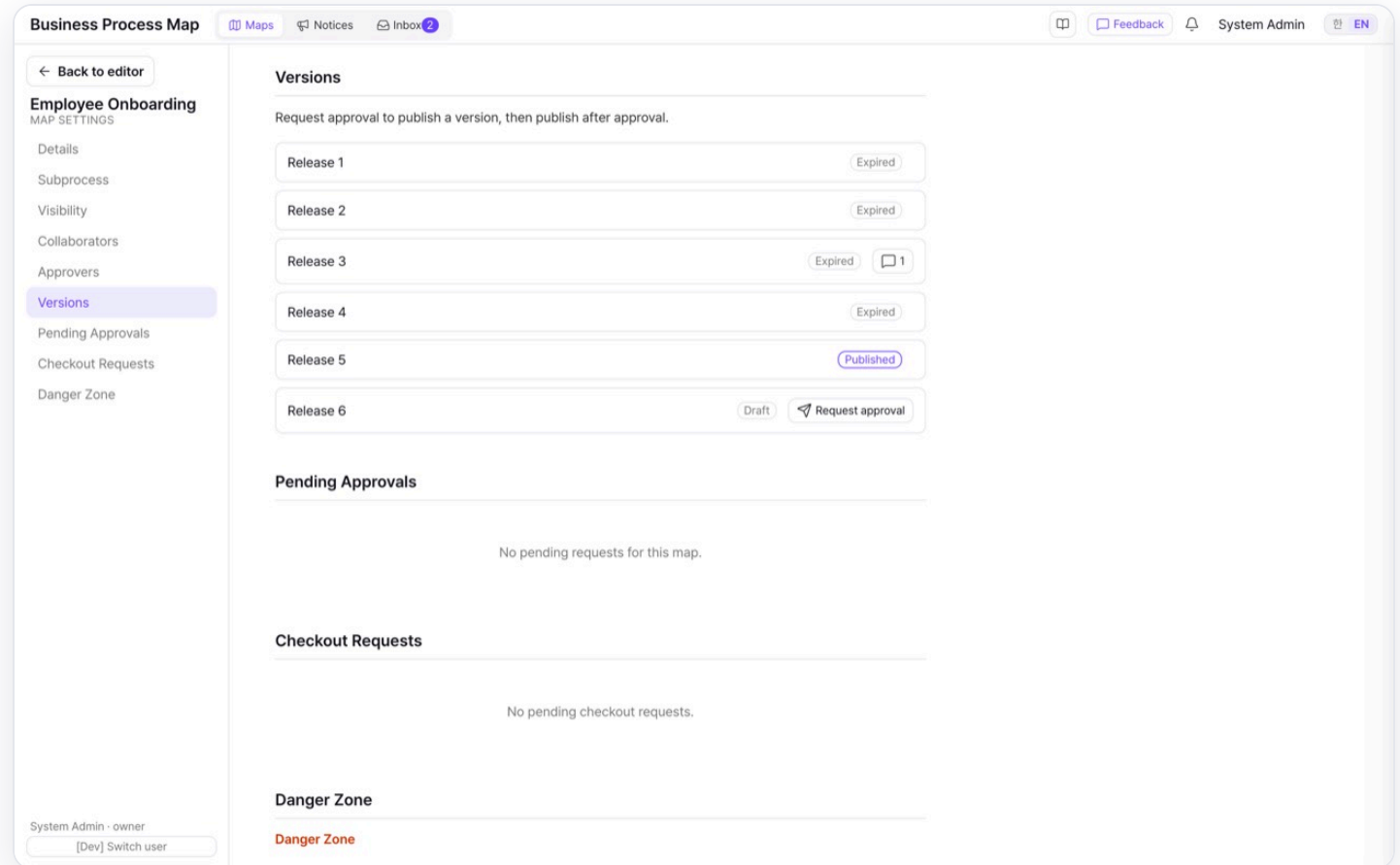
06

Versions & Approval

The draft → review → publish workflow, checkout, comments, and the inbox.

Versions & statuses

- A map holds versions (As-Is / To-Be / free labels) — a new one starts as a full copy.
- One draft at a time — a new version only after the current one publishes. You name it; the number is assigned at publish.
- Statuses — #Draft · #Pending · #Approved · #Published · #Rejected · #Expired (terminal).
- Publishing expires the previous one — Republish starts a fresh cycle from it.
- Framework linkage canvases (L5) use "Confirmed" instead of publishing — approval requests, publishing, approvers and collaborators don't apply (☹).



Checkout (single writer)

- A draft is checked out to one person — everyone else sees a "{name} is editing" read-only banner.
- Request / transfer checkout — the holder (or the map owner / a sysadmin) approves from the Inbox or hands it over directly.
- There is no time-based auto-release — it changes only by transfer, an approved request, or a sysadmin takeover.
- On rejection the checkout returns to the submitter — nobody can grab it in between.

From draft to published

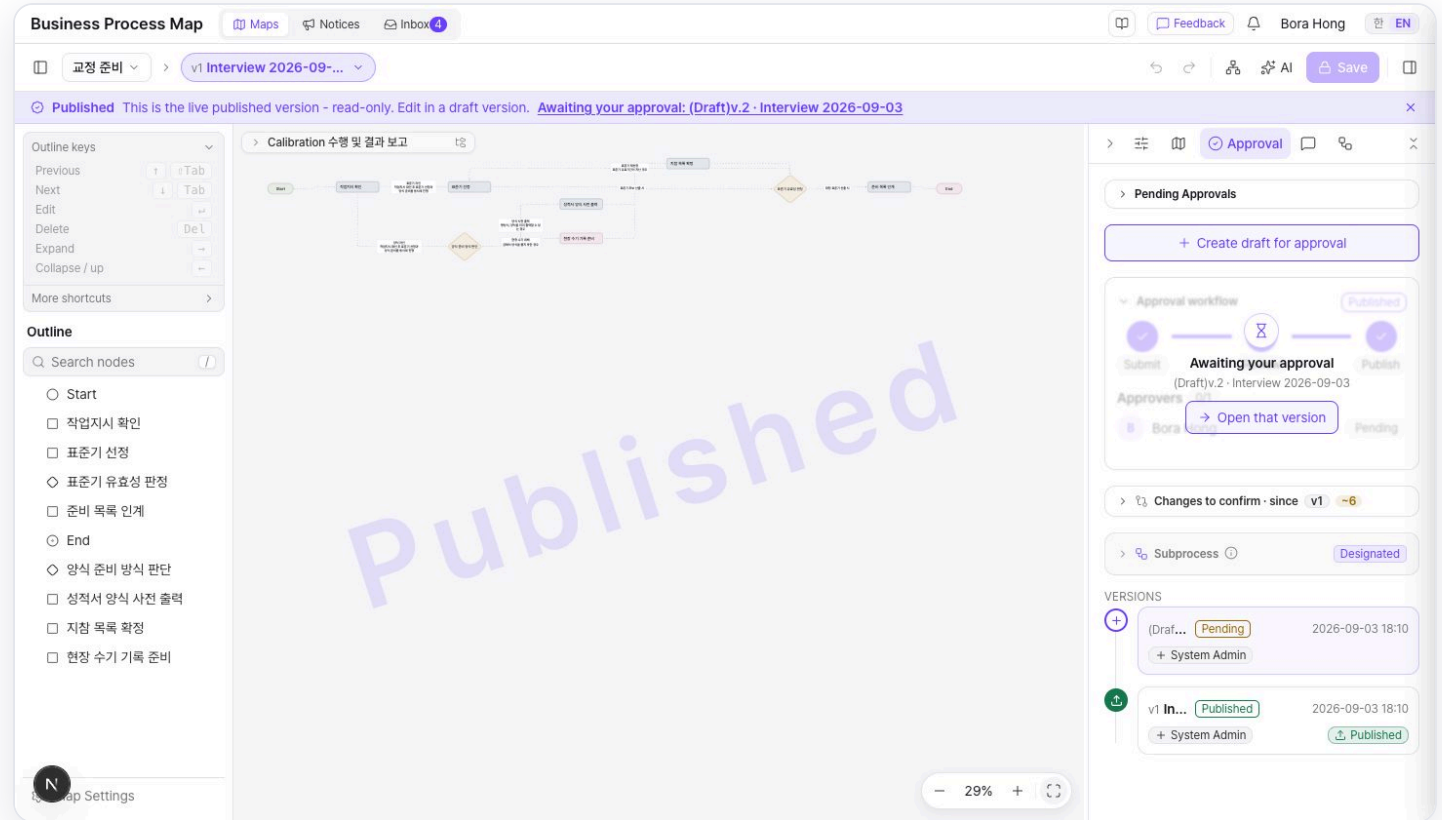
- ① Request approval — check the approver list, add a comment (optional). On an AI-enabled server the **AI draft** button writes a 2–4 line change rationale against the latest published version (the approver's AI summary reads it). The last rejection reason shows as a banner.
- ② Approvers approve/reject — rejections need a reason; approval is unanimous.
- ③ Publish — **only the submitter** can publish; the previous published version expires.
- Withdraw — Pending/Approved by the submitter only; Rejected also by owner/admin. The withdrawer gets the checkout.
- Self-publish — when you're the only approver, a "You are the only approver — Approve and publish in one go?" Yes/No popover runs the chain.
- Bundle visibility (owner) — attach a visibility change to the submission; it applies at publish.

Comments & history

- Comments from the submit/approve/publish/withdraw dialogs are **stored with the version history**.
- Versions with comments show a **speech-bubble count button** opening the history modal.
- Approval tab — per-approver tag pills (hover for time & comment), progress (n/m), and waiting-on pills.
- **Change summary** — under the approvers, the delta vs the published version (nodes & edges), collapsible.
- Opening a rejected version shows the **rejection banner** (who, why).

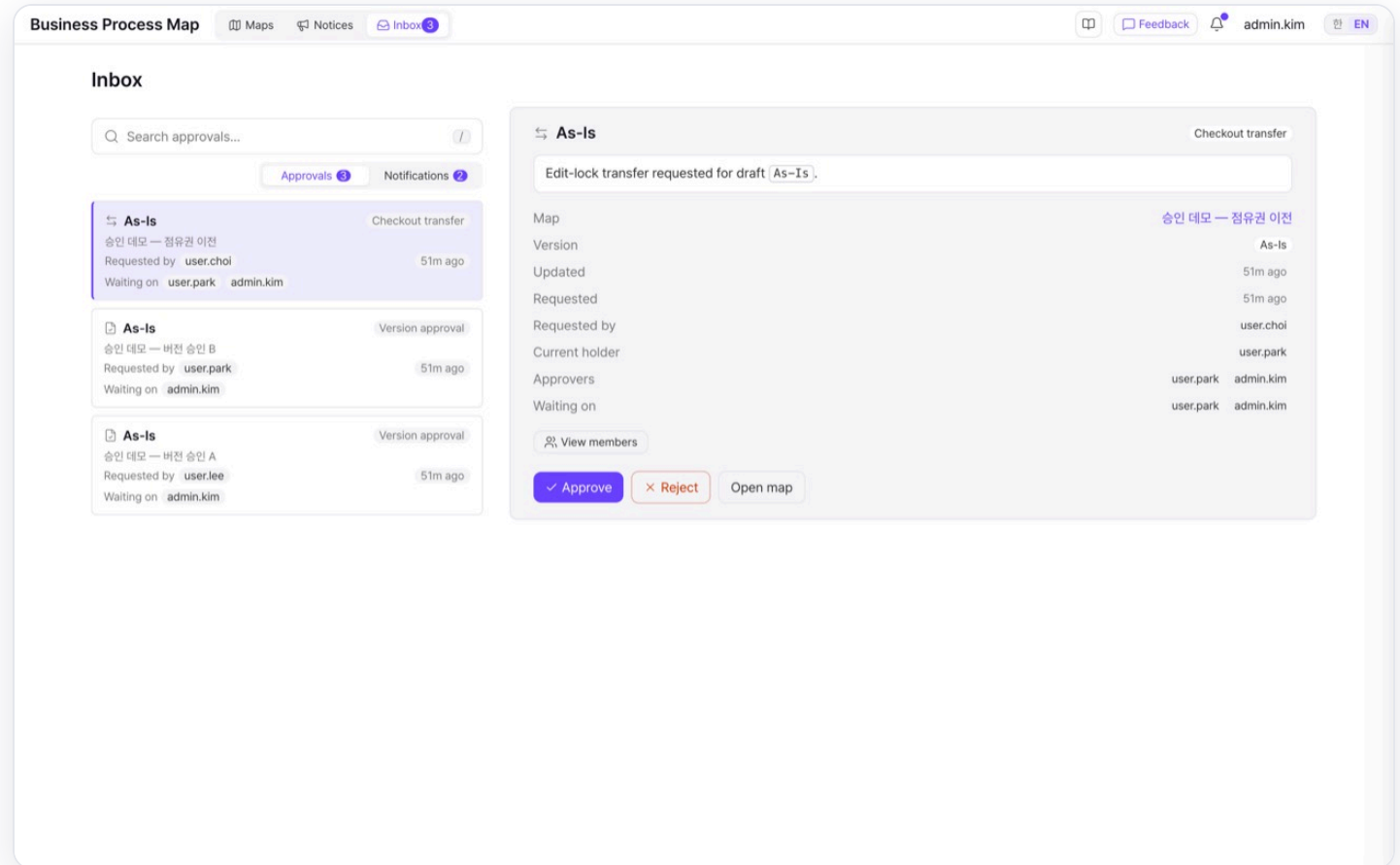
Approver landing & Compare with published

- When a **pending** version awaits your decision, opening the map **lands on it** (an explicit `? version=` still wins).
- On another version: an **Awaiting your approval: {version}** link in the banner and an **Awaiting your approval — Open that version** overlay on the workflow section (queue and designation card stay usable).
- On the pending version, the Approval tab's **Compare with published** opens latest published ↔ pending in the compare screen (with the AI summary tab).
- The overlay guards against deciding on the wrong version — approve or reject after opening that version.



The approvals inbox

- Your review queue — version approvals, checkout transfers, permission/visibility, renames, SP registration requests, Framework **confirm requests** and **slot changes**.
- Approve/reject inline (with a reason); Open map jumps to the source. The top Inbox tab carries your pending-count badge.
- SP registration card — Designate & approve: saving the form approves. Maps without a published version can't be designated yet.
- **Confirm request card** (direct L5 admin · sysadmin) — approving confirms the canvas on the spot; a canvas failing its gates can't be approved. Requesters may withdraw first.
- Requests bundled with a version show as read-only "Decided with version approval" rows.



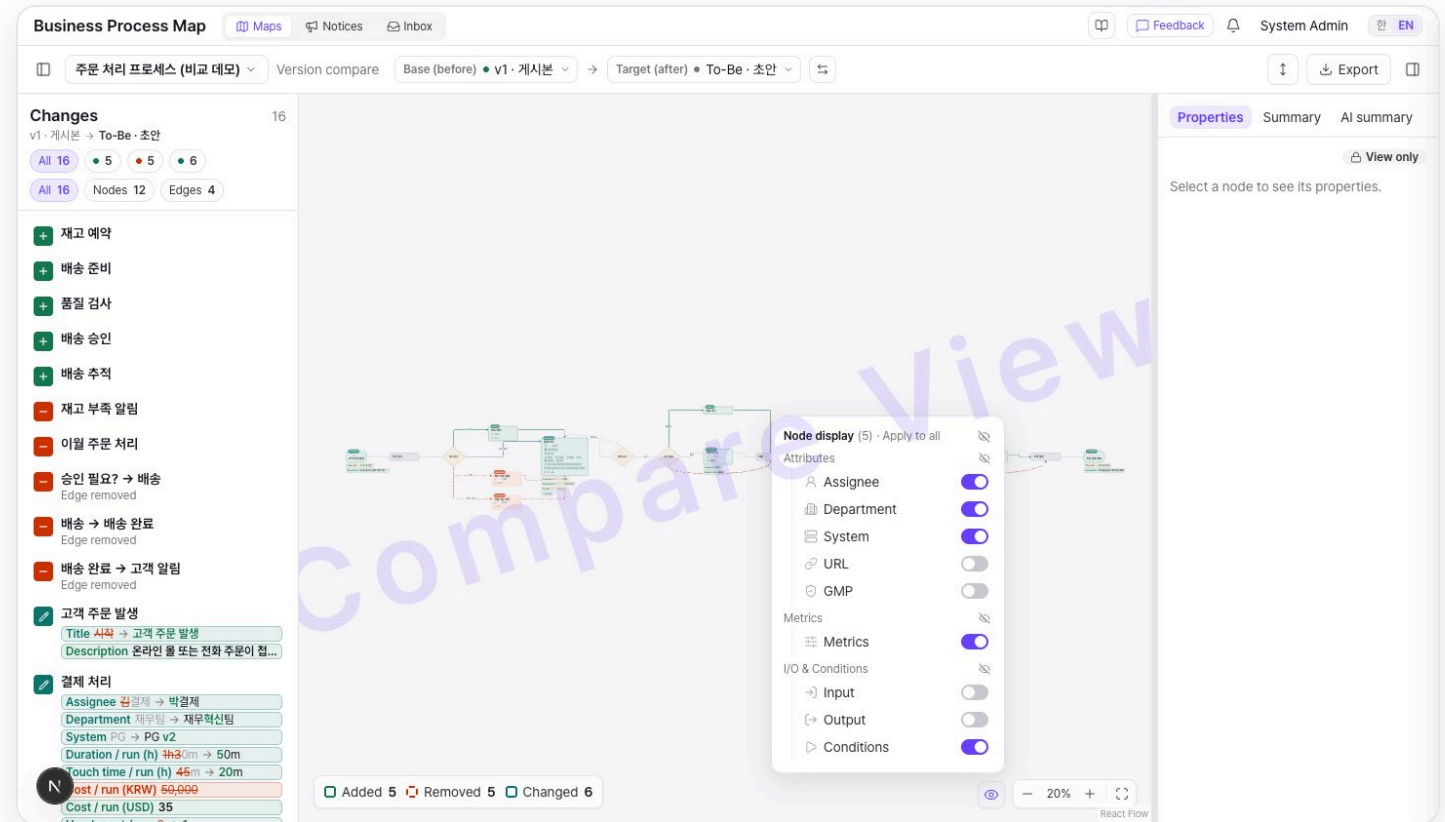
07

Comparing Versions

Read differences in color, and summarize them in numbers.

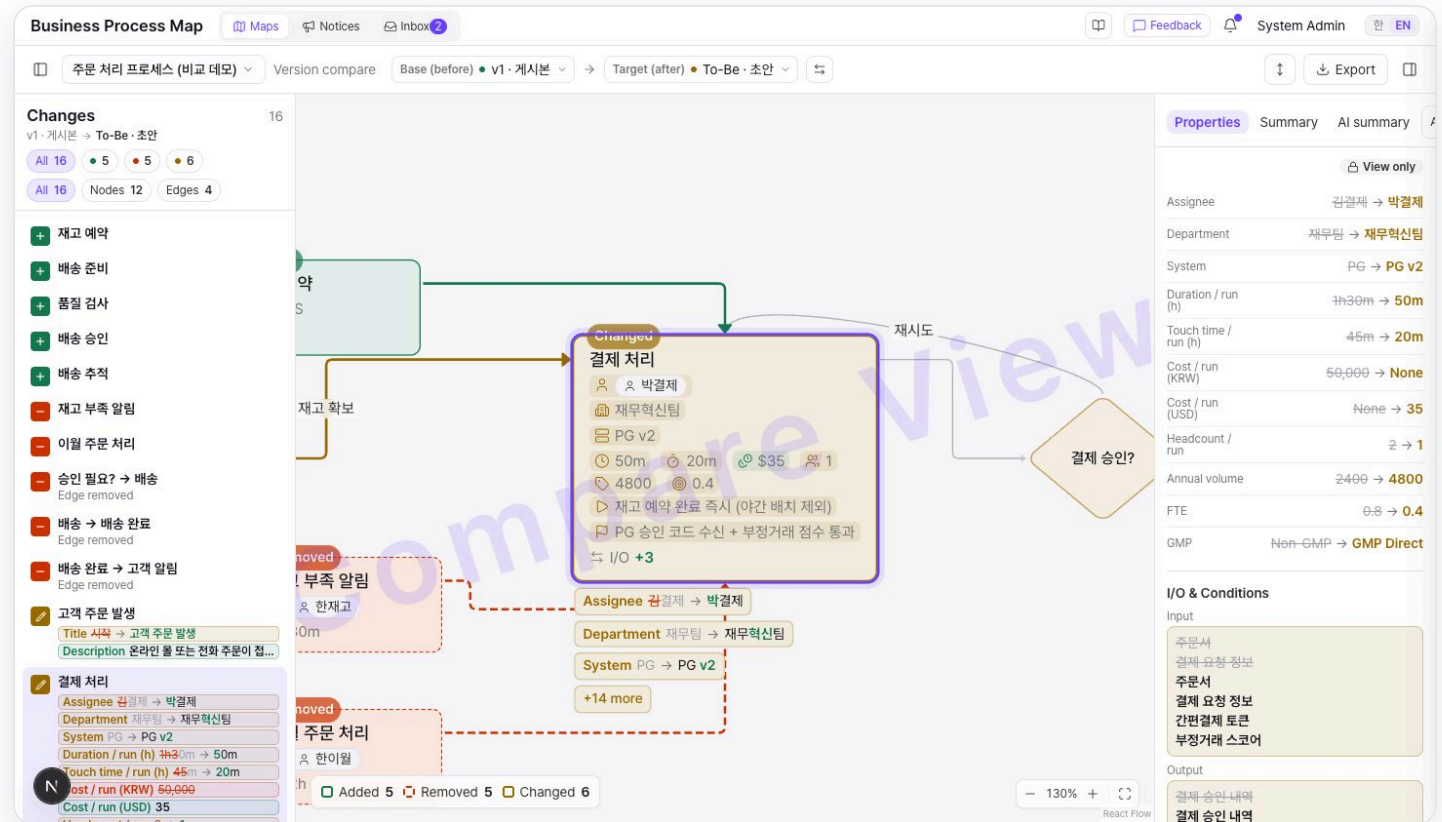
Opening & picking versions

- Open **Compare** from the editor top bar (needs a published version) or the Map tab's compare button.
- **Base (before) / Target (after)** captions make the direction explicit — echoed under the Changes panel.
- Dropdown rows carry a **status pill** and **last-updated date** — the other side's pick shows a tag; **click to swap**.
- Toggle horizontal/vertical layout; drag overlapping nodes **temporarily** (never saved).
- The first view opens with **base = published** and **target = the newest other version**. Drag the right panel's left edge (240–520 px); the **eye icon** next to the zoom pill picks the node display fields for the compare canvas.



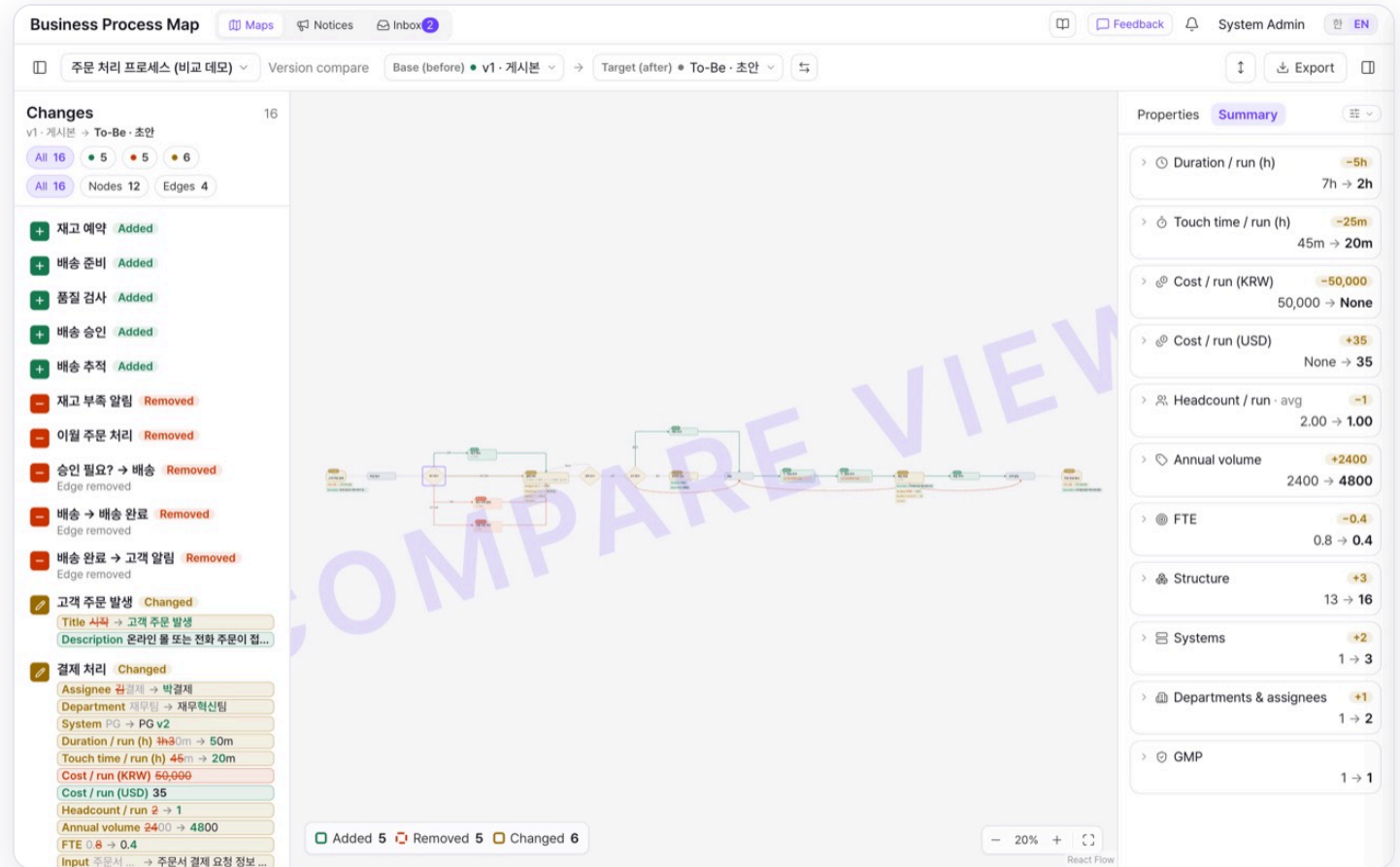
Reading the diff

- Added = green · Removed = red · Changed = yellow — edge-label changes too; long labels wrap.
- Per-field status colors — new green, removed red struck-through, changed with **only the changed part** emphasized; hover clipped values.
- Nodes carry role, department, system and metric chips plus conditions — I/O collapses to **I/O +N -M** (hover for per-item tooltip); changed fields get a status icon and tint.
- Properties pane **All / Changes only** toggle (default Changes only) — added, removed and unchanged nodes are dimmed under a watermark.
- Filter the change list by All/Nodes/Edges; click to focus. **Apply To-Be · Export** (PNG with info card).



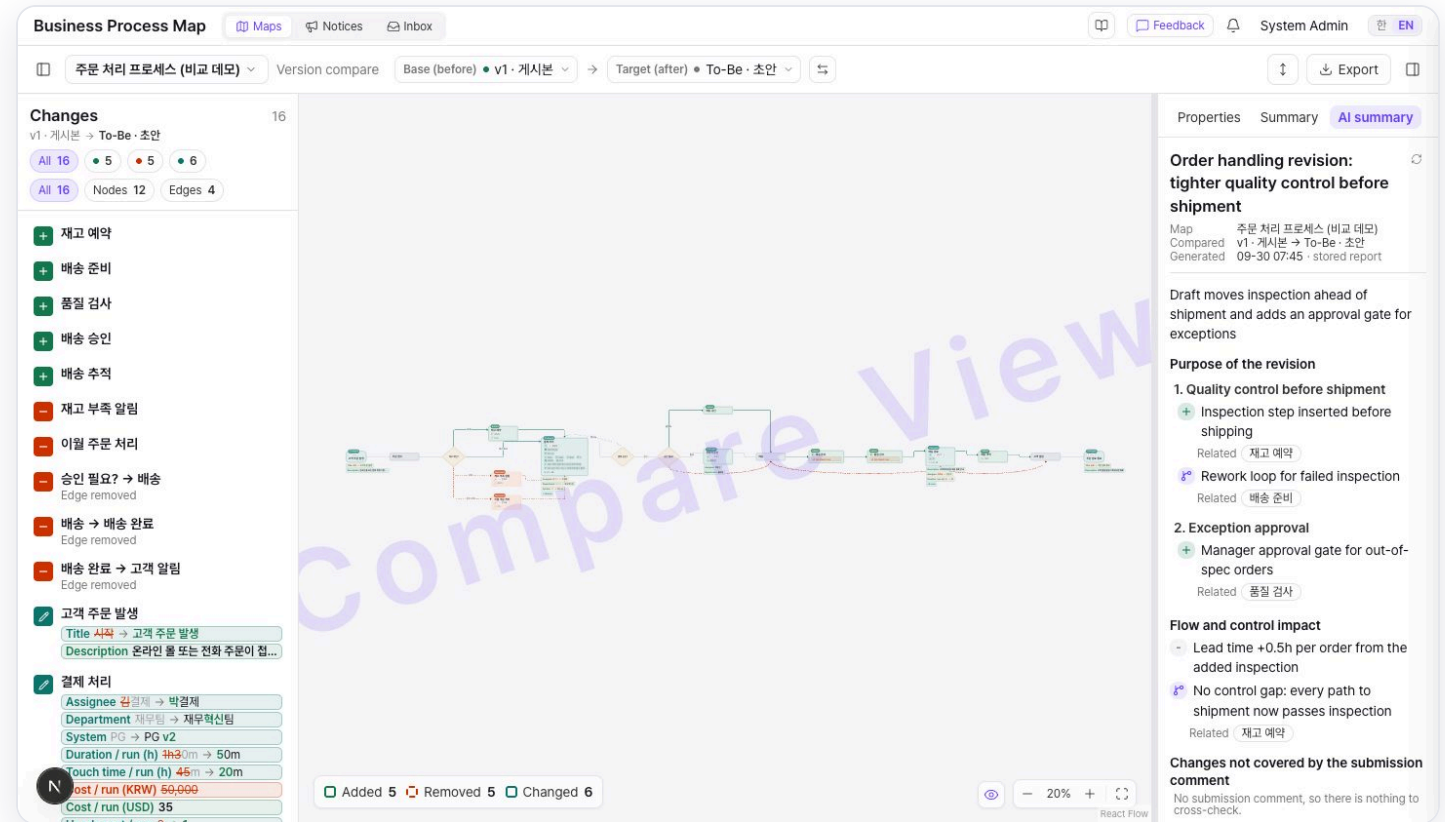
The Summary tab

- **Version totals** of the seven metrics (base → target with deltas) plus contributing nodes (click to focus).
- **Structure** (node/edge counts), **Systems**, **Departments & assignees** (coverage), and **GMP** distribution.
- Pick visible sections from the **Summary items** dropdown.



The AI summary tab

- The third tab, **AI summary** (AI-enabled server) — opening it writes a **bullet-style report for approvers** (spinner on the tab label).
- Four blocks — **Revision summary** (sections per business intent, related chips focus the canvas) · **Flow & control impact** · **Changes not mentioned in the submission comment** · **Questions before approval**.
- The server keeps the report — other approvers or a reload get the **saved report** at once; further draft edits regenerate it. **Regenerate** in the title row discards the saved copy.
- Open a specific pair via `?base=&target=` — the Approval tab's Compare with published uses this link.
- Very large diffs send at most 200 nodes and 200 edges to the AI.



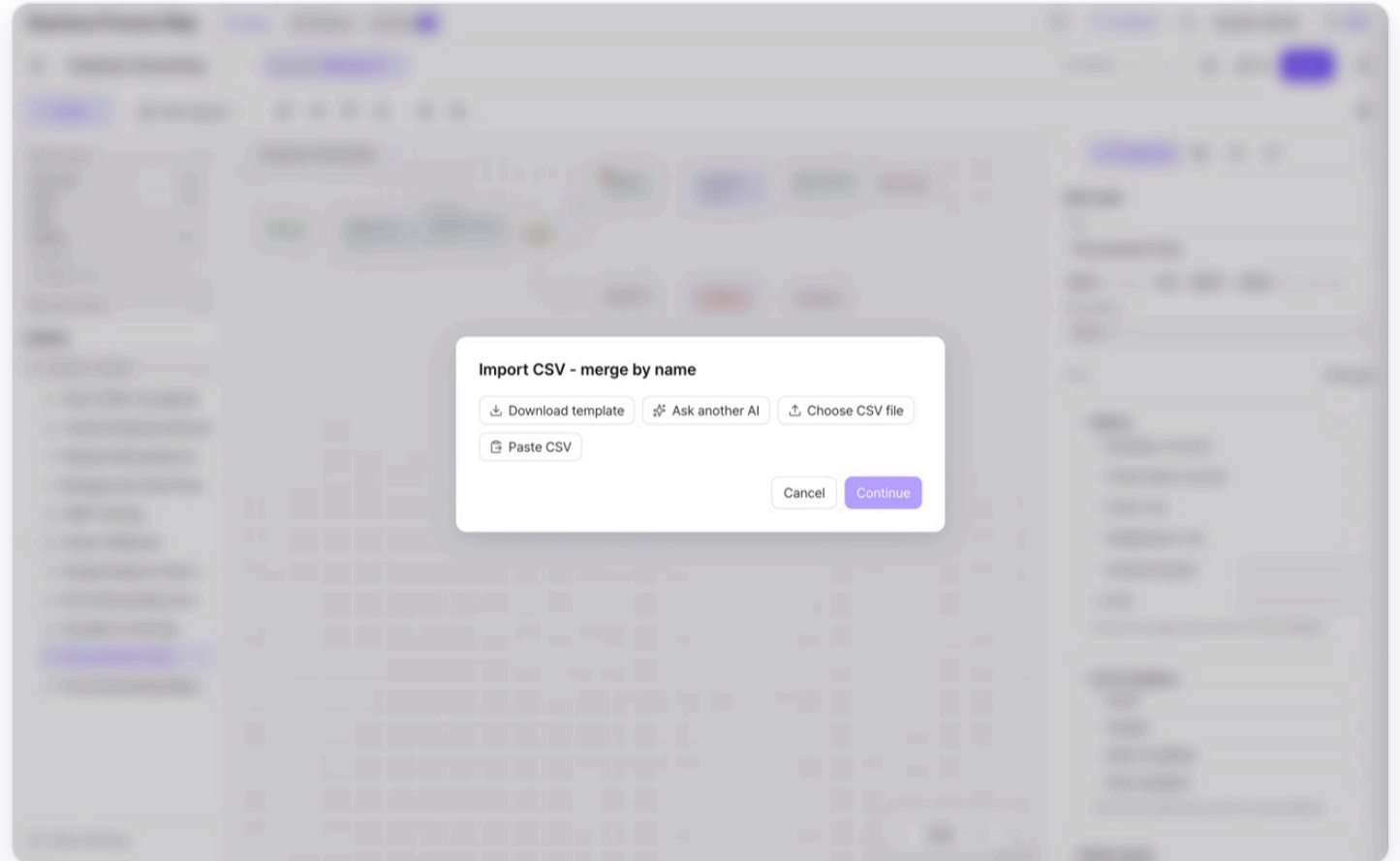
08

Import · Export · AI

Fill from CSV, export **four ways** (PNG · Excel · CSV · Word), and draw with the AI assistant.

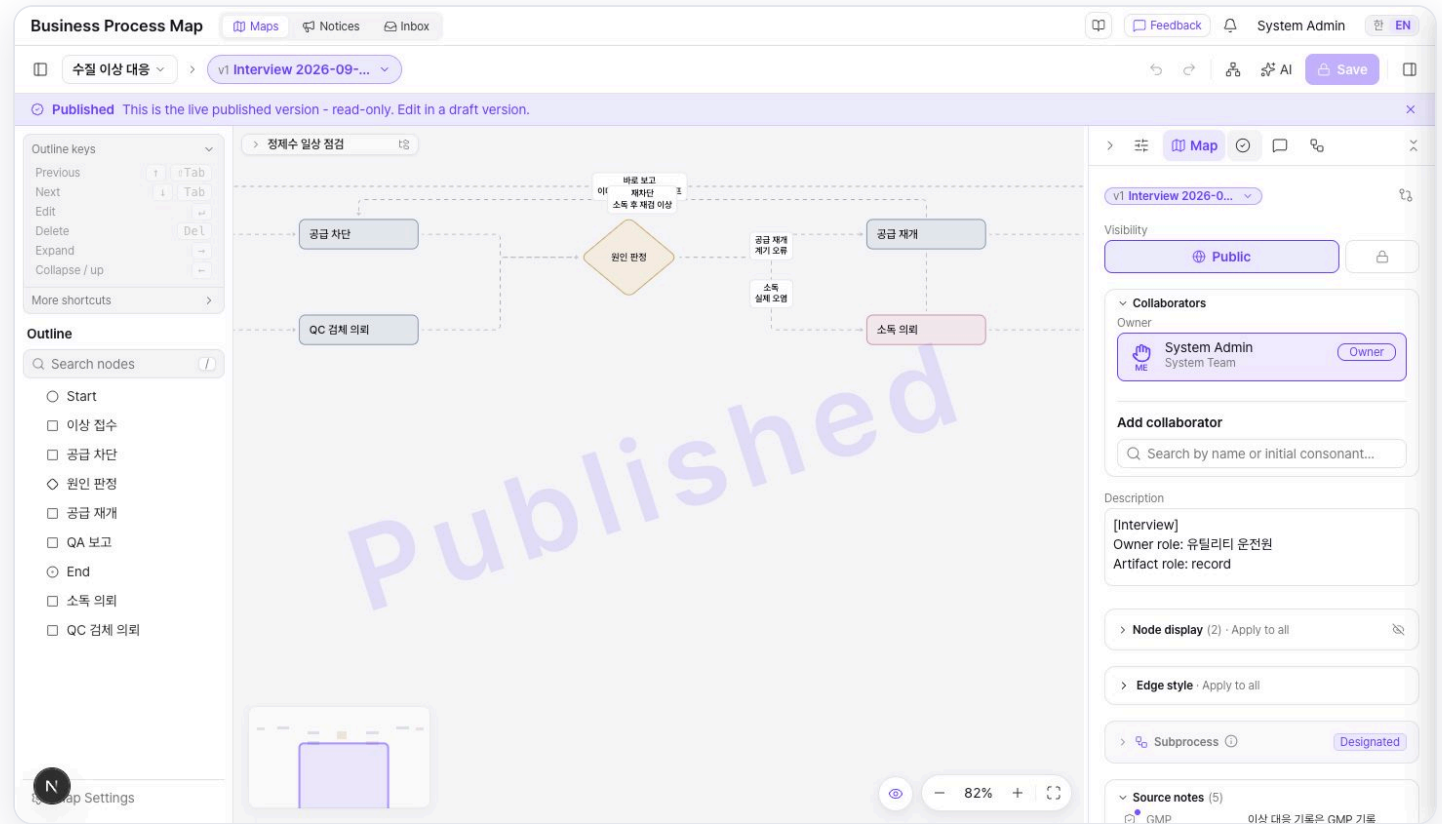
CSV import

- **Import CSV** in the top bar — download the template, fill, upload. **21 columns** (name required; `role` after `assignee`; attributes, metrics, I/O, `input_flags`, URL, next), max **500 rows**.
- Columns outside the list (e.g. legacy `data_form`) fail with **Unknown column**. Merged by name — existing nodes keep color/comments/groups; blank cells keep values.
- Assignees match the directory by name; **role** aliases → canonical (unknown kept as typed); **system** aliases → canonical, unknown → **Other + source note**. One currency, h.mm apply.
- Review the Added/Matched/Removed summary and warnings, then apply. Export puts an **Other** node's note in the System cell, so it round-trips.
- Ask another AI — copy the prompt, build the CSV with an external AI, paste it back.



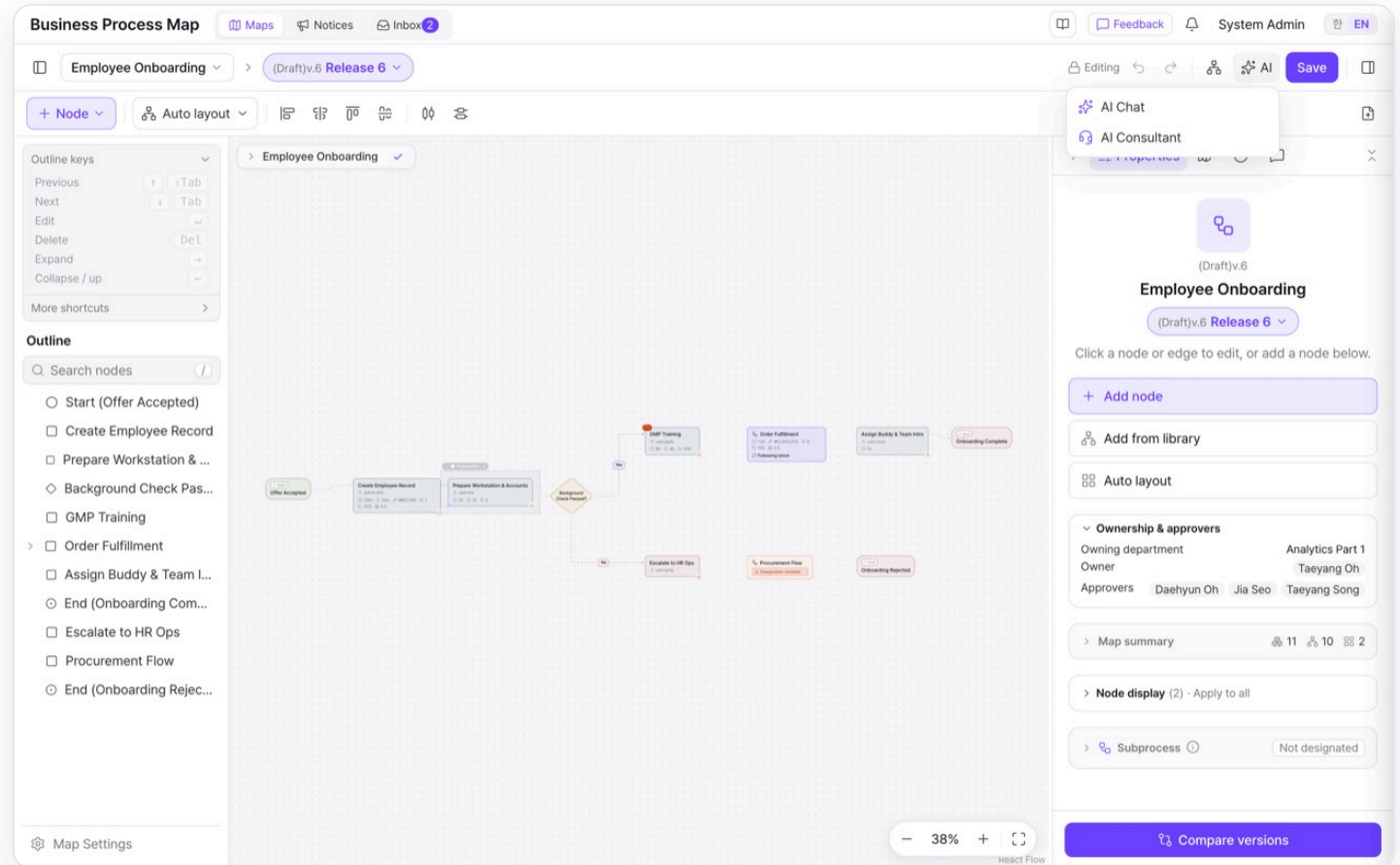
Export (PNG · Excel · CSV)

- PNG — 2x resolution with a map info card (name, owning department, owner, version, publish date, framework path); selection ring, I/O details and flow highlight are left out. Ctrl+Shift+E.
- Excel — **Process Map** or **WBS** (subprocess level columns), per-currency number formatting; truncated at 2,000 rows (and says so).
- CSV — the same 21-column table as import; round-trip by editing and re-importing (Other systems carry the source note in the System cell).
- CSV skips, with a warning, what a table can't hold — non-primary End nodes, edges into End, duplicate titles, ; / : in labels.



AI assistant

- Generate flowcharts from plain language — attributes and metrics filled in. **The AI fills roles, never assignee names** (pick people in the assignee picker); systems are normalized to the catalog.
- Incremental edits — layout/colors/groups preserved; nodes, links, inserts, branch labels, descriptions, I/O, conditions, URLs. Preview then Apply.
- Analyze (**Find issues**), summarize, walk through (prev/next/autoplay), **Suggest improvements**, and manual-grounded how-to answers.
- Chats are stored server-side and follow you — multiple chats, read-only access to other maps' chats.
- Ctrl+Enter sends · 2,000-char limit with a remaining ring · adjustable text size.



AI consultant interview

- A dedicated screen that fills an empty map through an **interview** — the **headset icon** in the top bar or the "Start with the AI consultant" balloon. Needs AI enabled and an editable draft.
- Left: the **working-map preview** (nodes show role, department, system and metric chips plus I/O and conditions; `Tab` / `Shift+Tab` steps along the flow); right: the chat panel. Click a node for **Ask about this**.
- The AI asks step by step; candidates appear as **choice buttons**. The **Roles & Systems** stage asks a role per activity (catalog candidates + free text) and **never asks for assignee names**. Unknown items can be **marked TBD and skipped**.
- **Attach documents (fast path)** — attach SOPs or work documents and confirm **Draw now** to draft the map directly.
- Every step leaves a **checkpoint** — rolling back folds the later steps into history.
- Finish with **review & apply collected parameters** and **merge into draft** (by title) — check out the draft in the editor first. **Start over** discards chat, attachments and the working map.

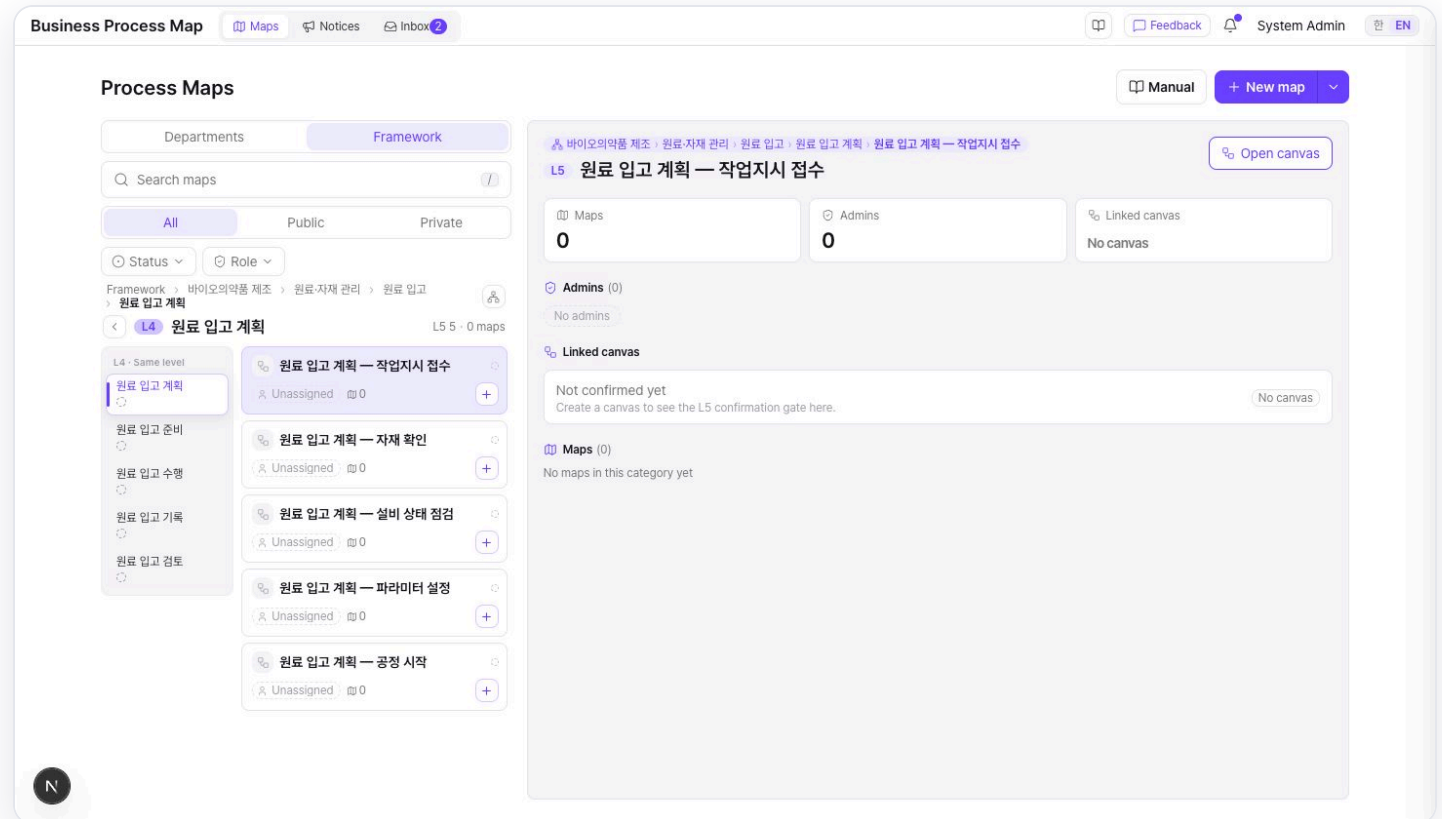
09

Framework & Linkage Canvas

Browse maps through the company framework (L1–L5) and draw cross-process linkage on L5 canvases.

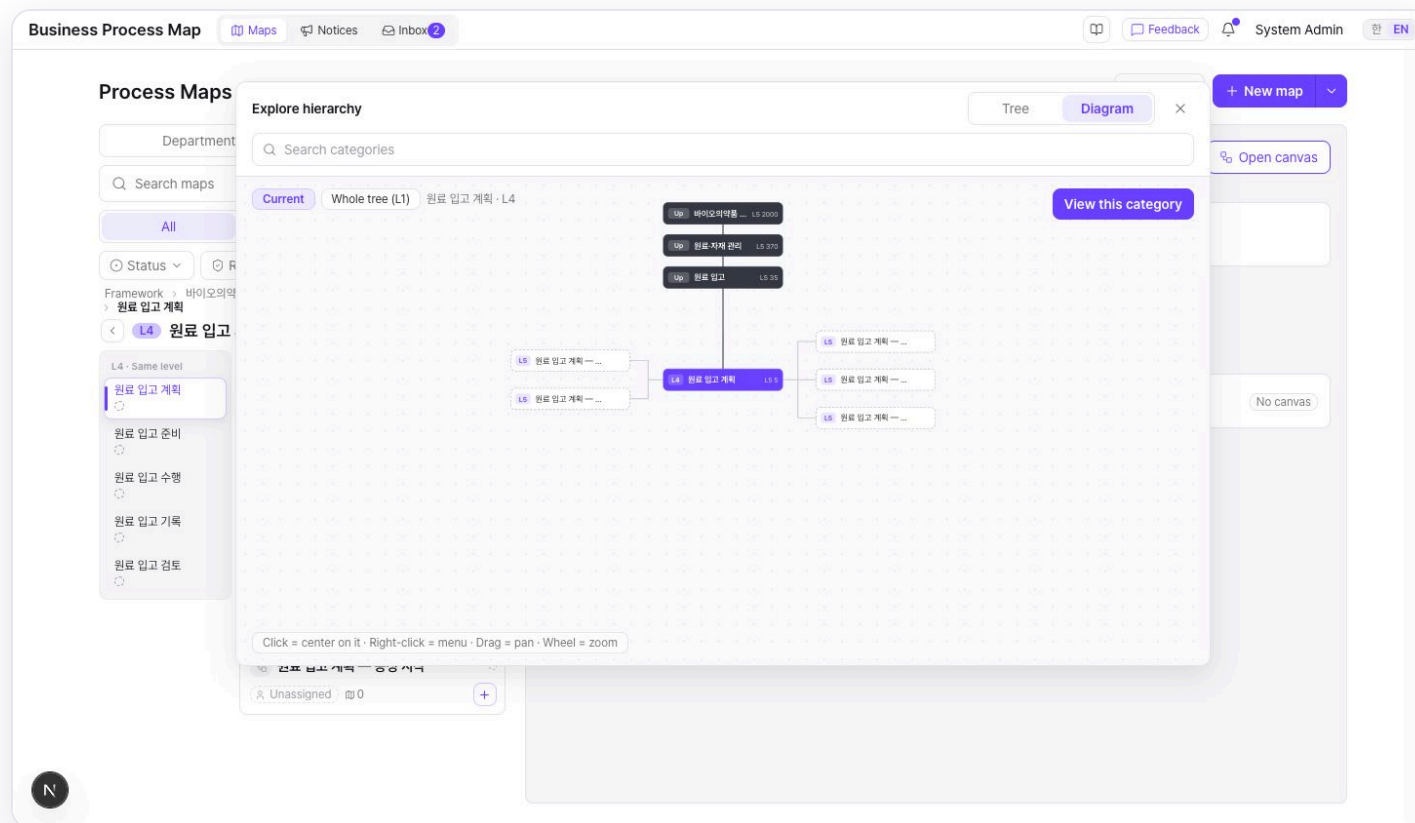
The Framework view

- The toggle above the home list switches **Departments / Framework** — the classification (L1–L5) explored **one level at a time** (pilot).
- Under the **breadcrumb: siblings | children** columns — a sibling swaps the right column, a child row goes one level down. Rows: level pill · name · direct admin · child count · map count.
- **L5s are cards** — state (Confirmed / Draft / No canvas) · admin · maps · Pending N · **Open / Create canvas**. Pick a card to list its maps in the summary panel.
- Maps attach to **leaf categories (L5)** only — assigning or transferring to a non-L5 is blocked.
- Search results group **by L5 category** (click the header to drill in; unregistered maps in a dashed group). From a map's detail, the **selected-map strip** returns to the L5. Your position survives reloads and searches.



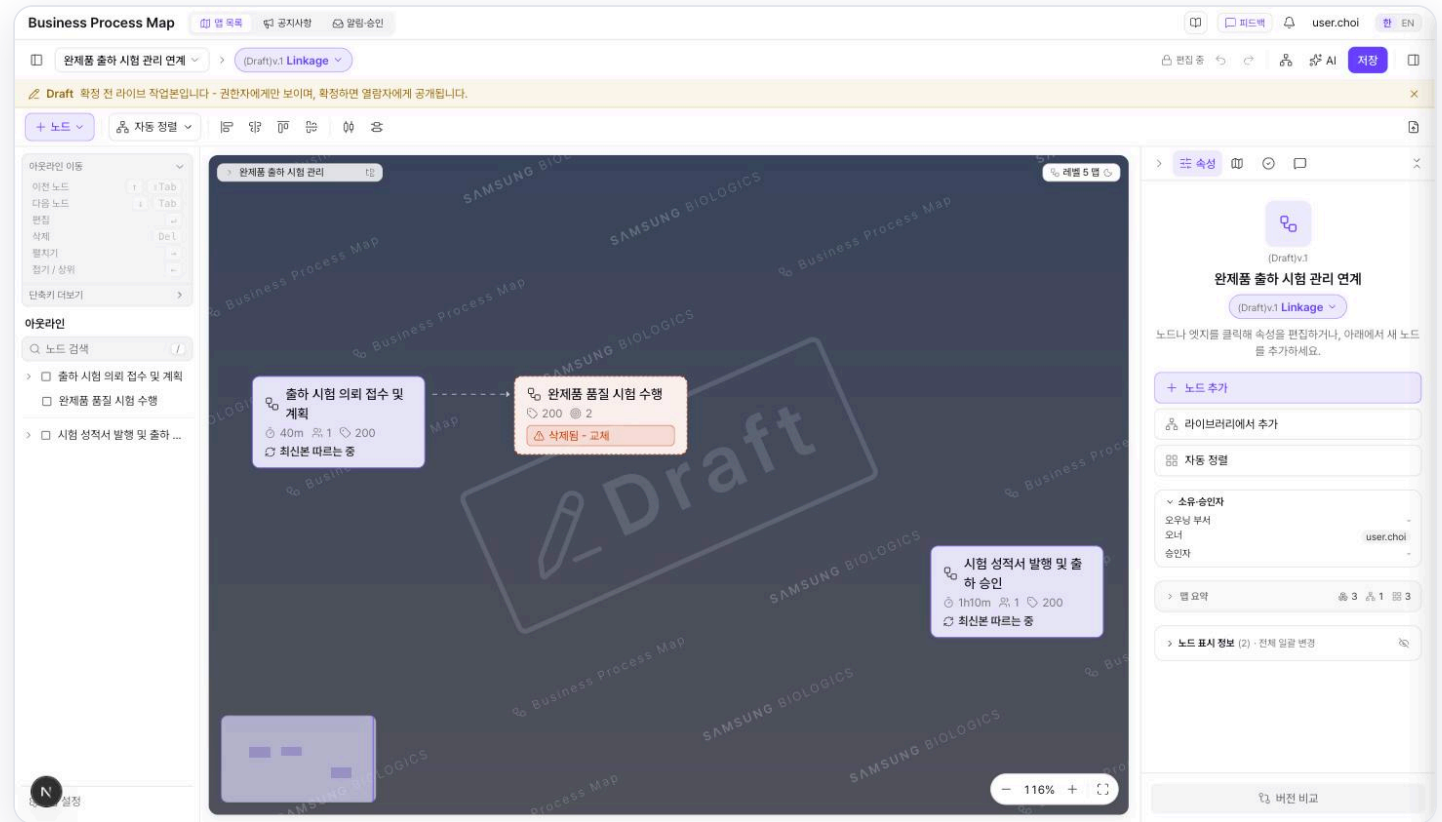
The hierarchy explorer

- The **tree icon** by the breadcrumb opens the **Explore hierarchy** floating panel — no dimming, draggable header, closes only with x/Esc, follows your drill-down.
- **Tree** tab — the current path pre-expanded; hovering a descendant highlights its ancestors.
- **Diagram** tab — ancestor chain → current → children → grandchildren, ERD-style. Click = center, **right-click menu** (Details · View this category · Center on this), **Whole tree (L1)** toggle.
- **Search** — substring and Korean initials (non-contiguous too); result rows show level pill (hover = ancestor path), name, L5 count; **Go** drills down.



The linkage canvas (L5)

- A special canvas with the category's processes (L6) laid out as subprocess nodes — no Start/End; decision and end nodes as helpers.
- A **charcoal background** by default — the button by the **Level 5 map** tag switches to light (remembered). Watermark plus a **Draft / Confirmed / Superseded** stamp.
- Category admins edit (higher grants inherit downward); the draft follows the single-writer checkout rule; new L6 maps auto-reconcile on edit entry.
- Contained L6 nodes cannot be deleted — the classification is the source. Regular map workflows (approval, publish, approvers, SP designation) are **absent**.
- **Notes** at the bottom of the Map tab belong to the category — entry/flow/open items from import, managed by admins and sysadmins.

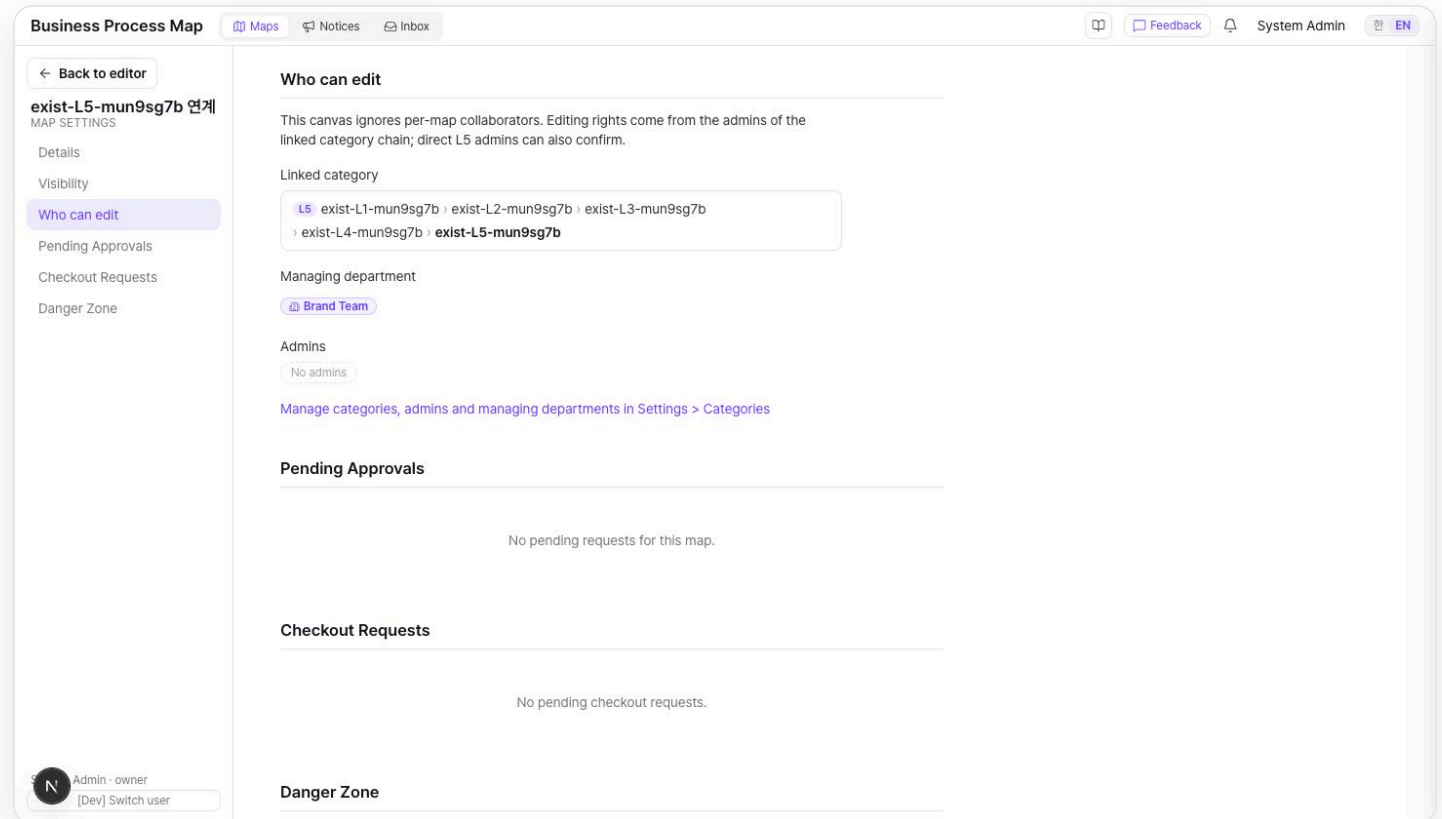


Who sees the draft

- Before confirmation the **live draft** is visible only to admins of the category (or a parent) and sysadmins.
- Everyone else lands on the **latest confirmed snapshot** — with no snapshot yet they see a "not confirmed yet" notice.
- A status banner sits on top — **Draft** (amber: pre-confirmation work), **Confirmed** (latest snapshot, read-only), **Superseded** (an older snapshot). Dismissable; returns on re-entry or version change.
- If a slot change re-points the canvas server-side, saving the old state is blocked with "**the canvas changed on the server — reload**".
- The version timeline groups snapshots by major and labels the newest **Latest confirmed**.

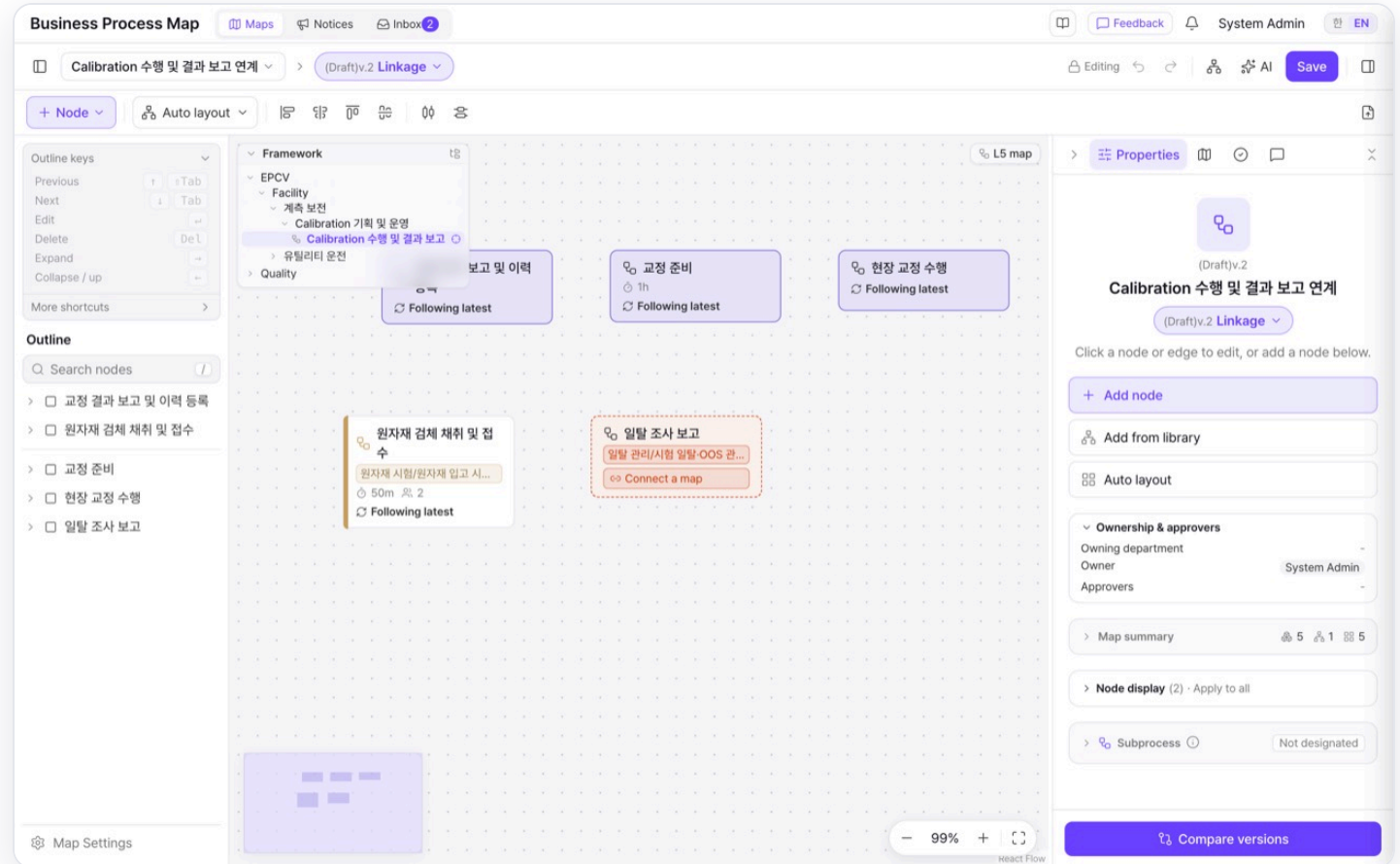
Linkage canvas settings — Who can edit

- A linkage canvas (L5) has only **Details · Who can edit · Visibility · Pending approvals · Checkout requests · Danger zone** — no Collaborators, Approvers, Versions or Subprocess designation tabs.
- **Who can edit** (read-only) — the linked category path, the managing department and the admins along the category chain; direct L5 admins carry a **can confirm** mark.
- To change who edits, adjust category admins or the managing department under Settings → Framework.
- Pending approvals collects this canvas's confirmation and slot-change requests.



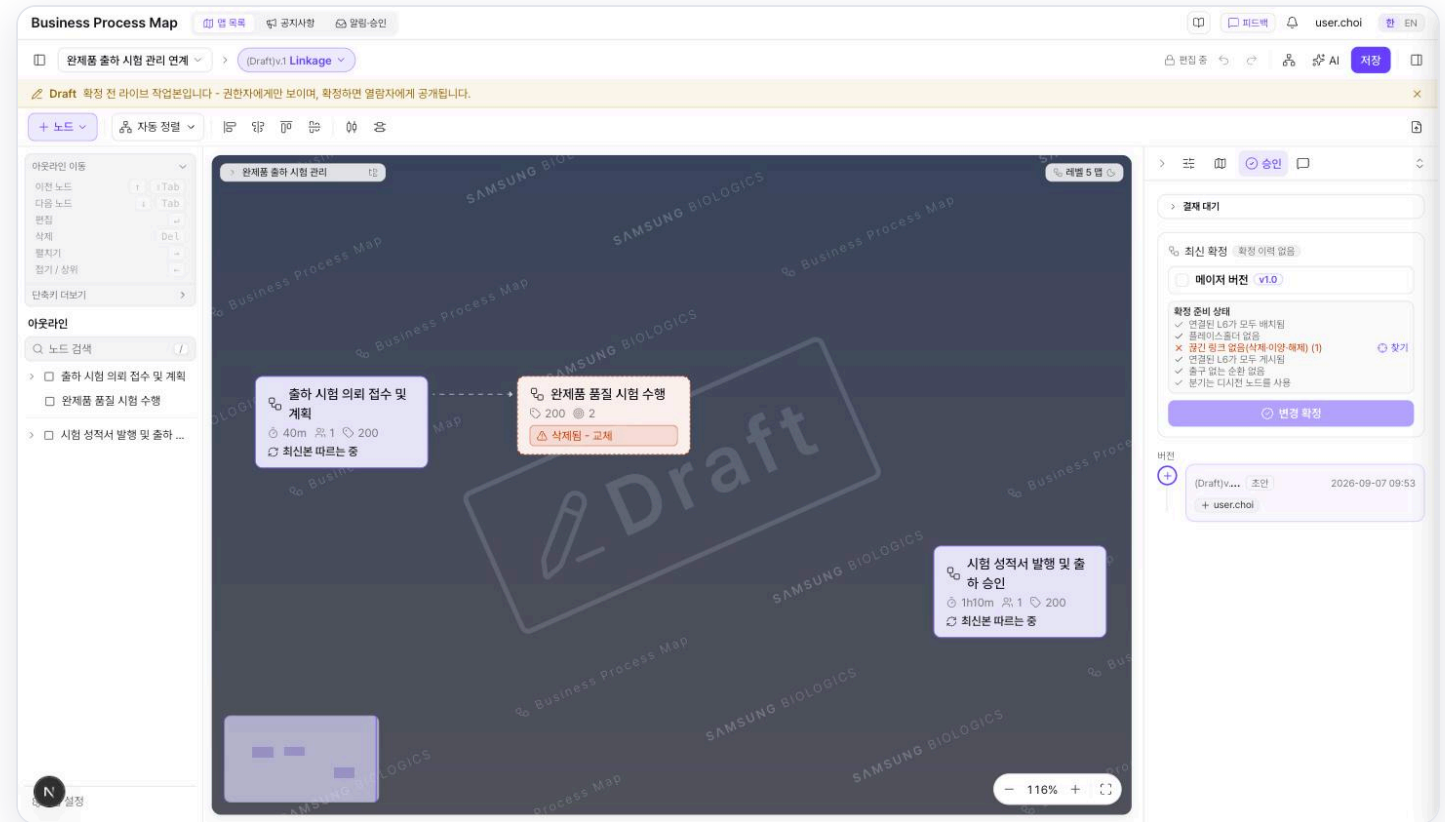
External L6 & the L5 explorer

- The top-left **L5 explorer** — see where you are in the tree and move to another L5's linkage canvas.
- Drag another category's L6 card in from the **Framework L6 picker** (library button · right-click **Add L6 task** · S) — rendered neutral with a colored left tab and origin badge.
- Click the origin badge for that L5's drill-in peek; "Search other frameworks" opens the browse modal (search & travel, gated by confirmation).



Confirm & versions

- **Confirm readiness (six gates)** — all L6 placed · no placeholders · no stale links · all L6 published · no exit-less loops · branches use a decision. **Find** jumps to the offending node.
- **Who confirms** — the checkout holder, if they are the L5's **direct admin** (or a sysadmin). Higher-level admins send a **confirm request** (checkout auto-releases) — withdrawable, answered by notification.
- Confirm changes — v1.0 → v1.1 ...; check **Major** for v2.0 (only the previous major's **intermediate minors** are pruned; X.0 and the last minor stay — the dialog previews them).
- Layout-only moves can't confirm ("no changes since") — the section shows the change summary since the last confirm.
- Snapshots open read-only from the version dropdown and compare against each other.



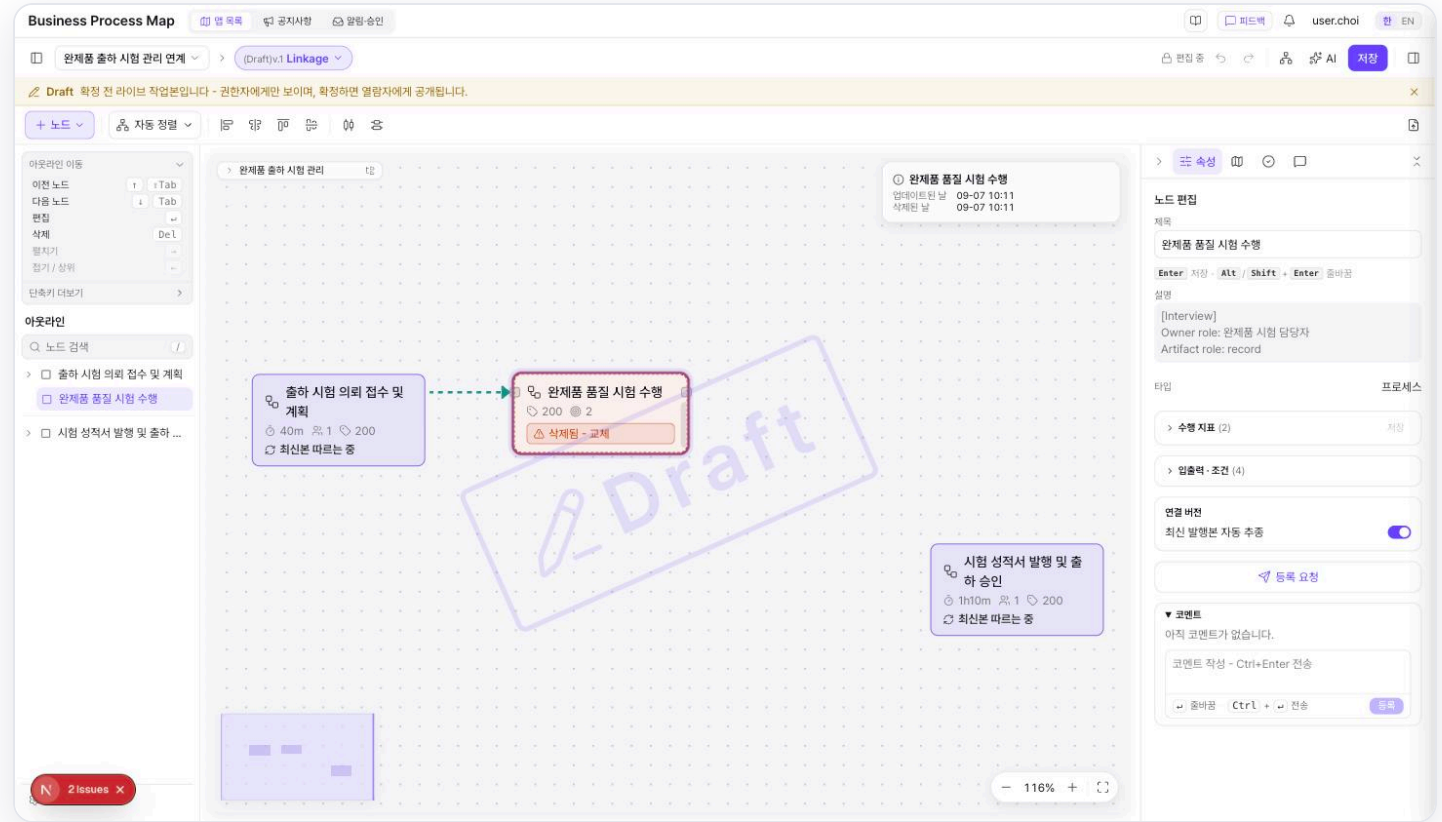
Placeholders & replacement

- The slot for an unregistered L6 renders as a **dashed, error-toned node** — the bottom banner's **Connect** opens the connect dialog.
- The dialog is a **framework tree** (auto-expanded to the guided L5) — click a map for the preview flyout on the right, then **Connect**.
- Maps outside the guided L5 ask for confirmation; maps already on the canvas show as **placed**.
- Editors can add a placeholder themselves via **Create placeholder** at the bottom of the L6 picker — the "no placeholders" gate holds until it is connected.



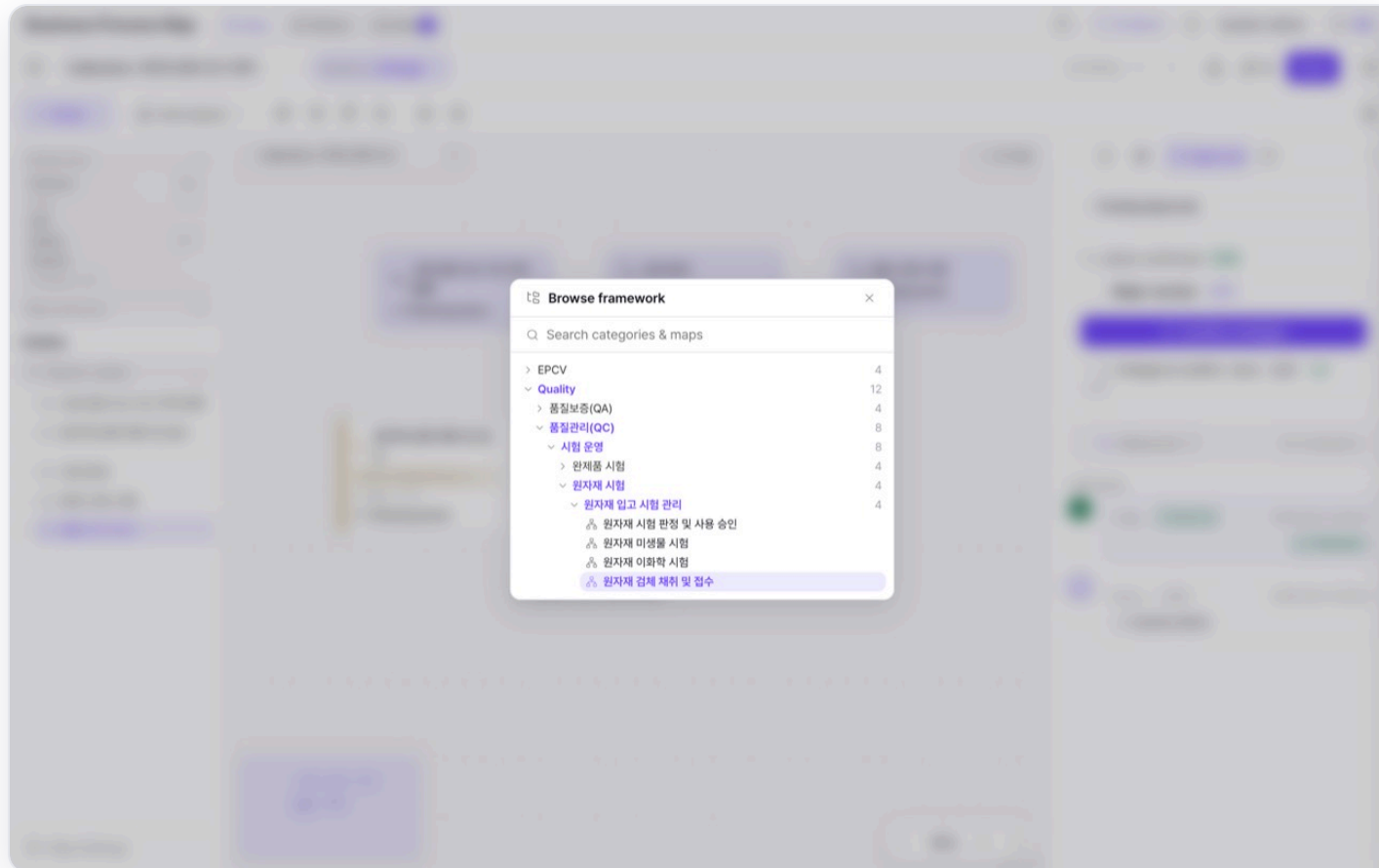
Replacing stale links & slot history

- Unassigned from the framework → **"Removed from the framework - replace"**; slot handed over → **"Handed over to (map) - replace"**; deleted with no successor → **"Deleted - replace"**.
- The node turns dashed/error-toned and the **stale links** gate on the confirm checklist catches it.
- Click the banner for the connect dialog — the **successor map is pinned as a recommendation card** for a one-click replacement.
- A node that just took over a slot wears a **"Recently handed over"** badge for 14 days.
- Hover an L6 node for its **slot history** (assign · move · unassign · delete · hand-over) with timestamps, top-right where the Level 5 map tag sits.



Drill-in peek & browsing

- Origin badges on external L6s and folder pills on SP nodes open the **classification peek**.
- The peek's "**Search other frameworks**" — a tree opened to the current path, with category/map search.
- Click a map to travel — leaving the editor passes a **confirmation gate**.



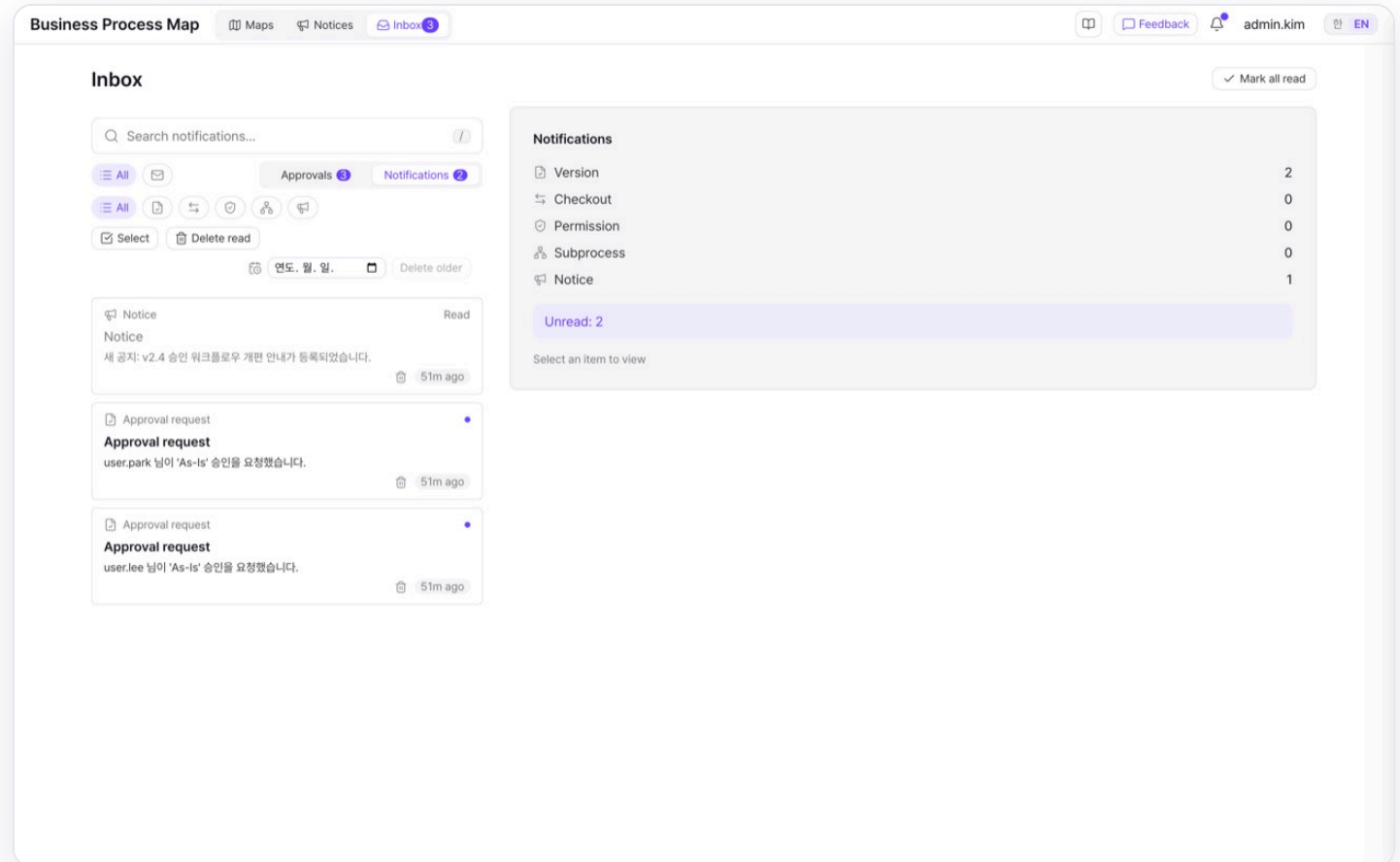
10

Notifications · Notices · Collaboration

Rich notifications, notices, feedback, and user groups.

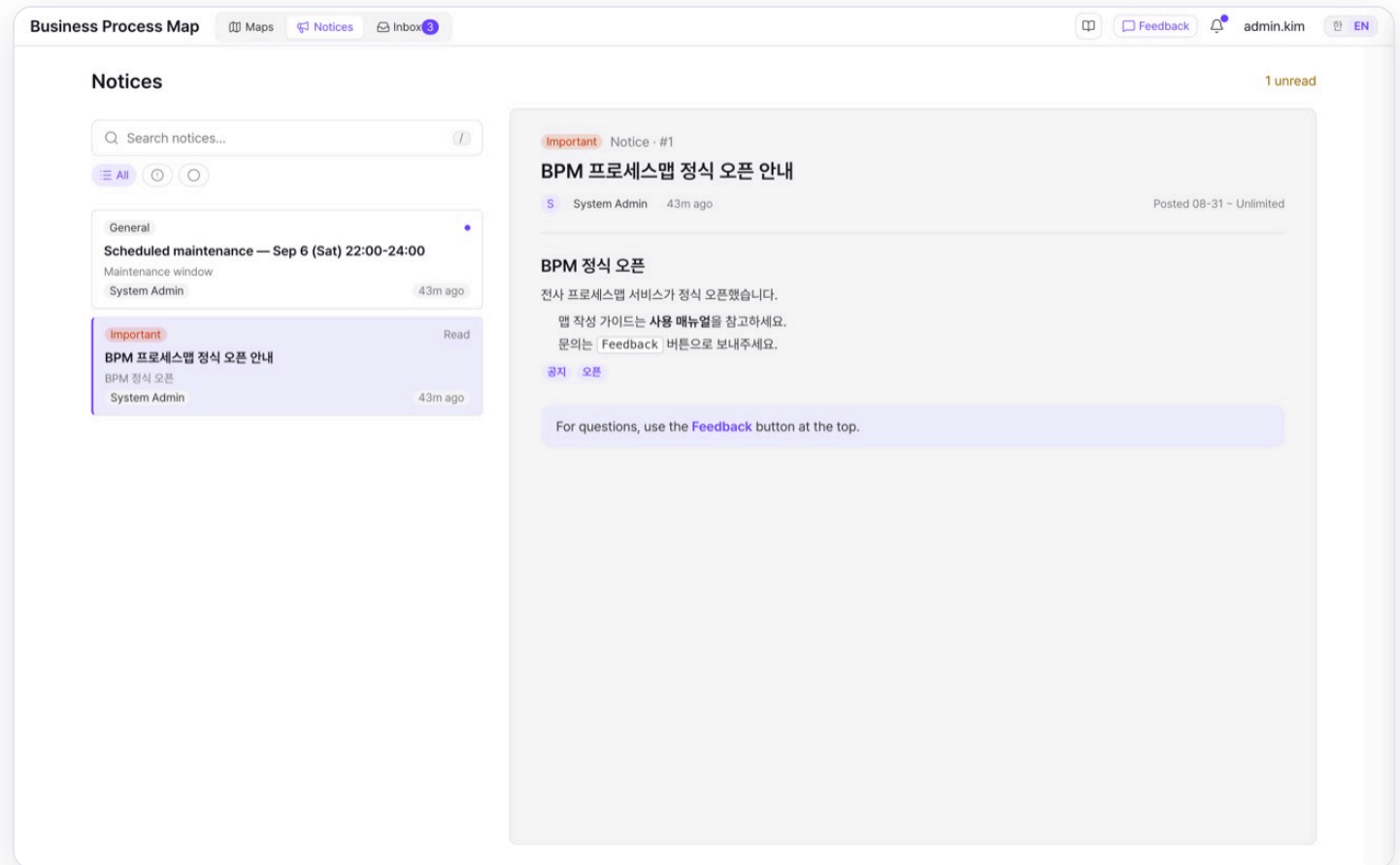
Notifications

- Sentences carry the **map name, version, and actor** in your language (older ones keep their original text).
- Hover/click the actor pill for the **person card**; the version chip opens a **version card** (status, time, jump).
- The **bell** refreshes every 5 s — clicking jumps to the tab and marks read. The tab has mark-all-read, category filters, and bulk deletes by selection/read/date (permanent).
- **Data clean-up** — when your map still references departments or people gone from the org chart, the admin audit sends one **Stale refs fix request** per owner (map list attached); an owner change sends **Owner assigned**.
- Retention is the **most recent 100 per person** — overflow auto-deletes oldest-first.



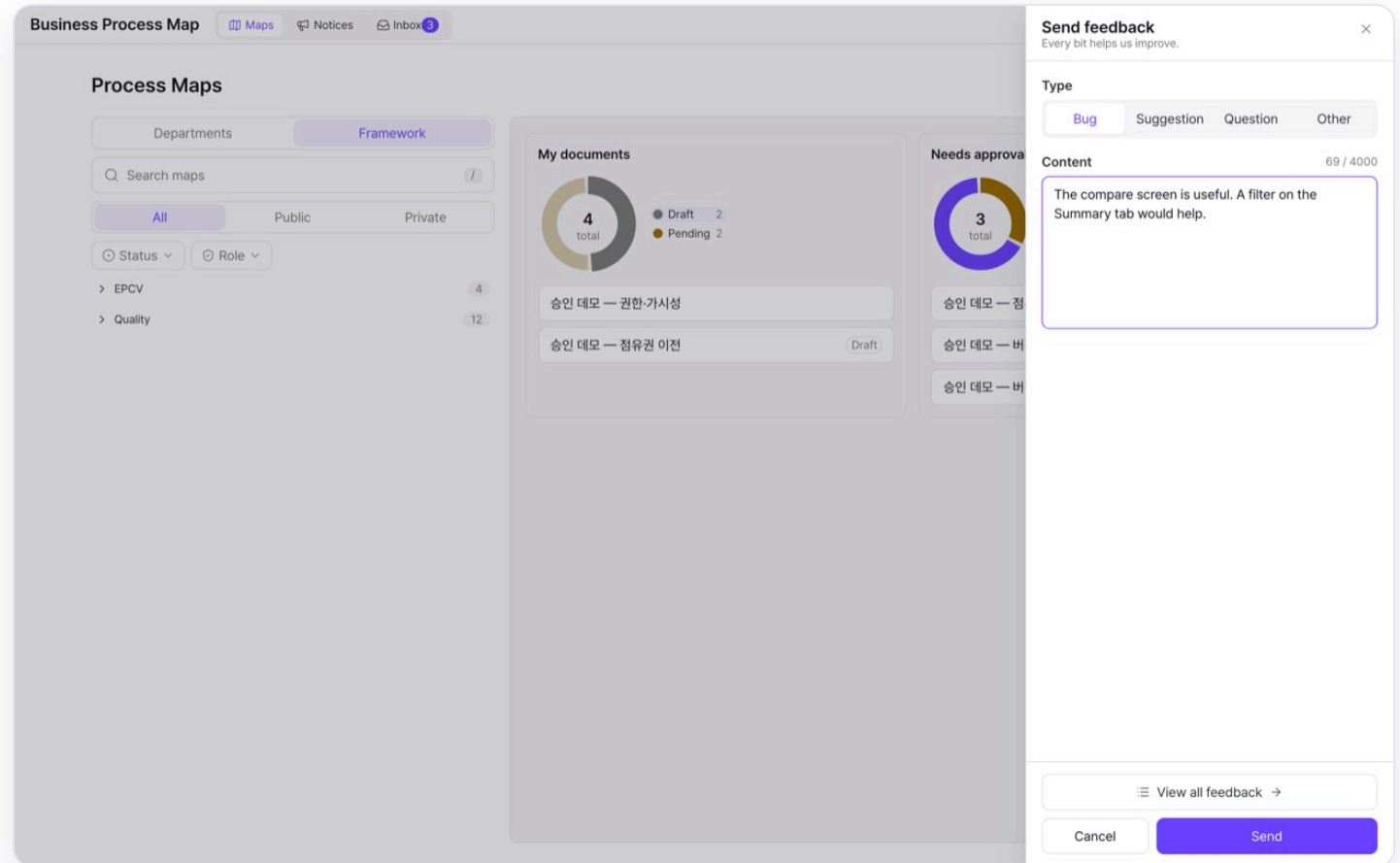
Notices & the manual

- The **Notices** tab — All/Important/General filters, search, Markdown bodies.
- The read mark is stored **only in this browser (device)**.
- **Manual** — from the top bar, with a document dropdown (KO/EN) and searchable TOC/body.



Feedback & user groups

- Feedback — Bug/Suggestion/Question/Other with your screen and open map attached; track status and replies on the Feedback page.
- Notes — anyone can add running memos; edits keep history, deletion archives. Reply/status notifications are sent deliberately by an admin.
- User groups — request with 2+ members and 1+ manager; active after admin approval. Grant map access in one step.
- Managers — add/remove members, appoint managers, rename (once a week), deactivate/reactivate, delete (7-day trash). You see only groups you belong to or manage.



Quick FAQ

- Can't edit? Check the banner — checkout, approval in progress, or Viewer role.
- Can't create a version? Finish the current draft cycle through publish.
- Won't save? One Start, one primary End, unique End names, no multi-output plain nodes.
- No draft on the linkage canvas? Drafts are admin/sysadmin-only — everyone else lands on the latest confirmed snapshot.
- Can't submit? Assign at least one approver first.
- Copy disabled? Only maps published at least once can be copied.
- Linked subprocess locked? It's undesignated — request registration, or publish and designate.
- "Owner unconfirmed" on a card? An interview-imported map — clear it via Map Settings → Danger Zone → Transfer ownership.

Happy mapping

Questions? Use the in-app **Manual** and **Feedback** — admins, see the Administrator Manual.